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(54) **PHOTOGRAPHING OPTICAL LENS
ASSEMBLY, IMAGE CAPTURING DEVICE
AND ELECTRONIC DEVICE**

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G02B 9/64 (2006.01)

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CPC **G02B 13/0045** (2013.01); **G02B 9/64**
(2013.01)

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USPC 359/708
See application file for complete search history.

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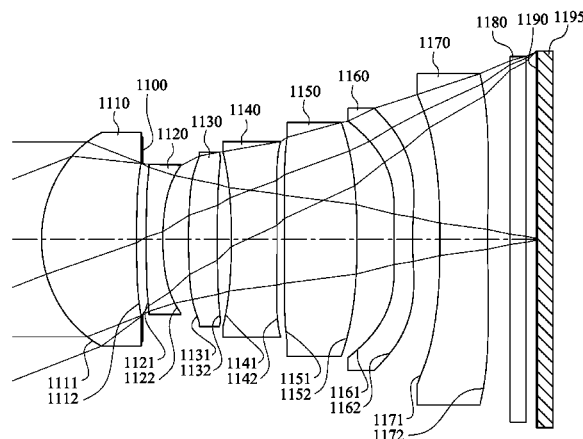
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Rodack, LLP

(57) **ABSTRACT**

A photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has negative refractive power. The sixth lens element has at least one of an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element comprises at least one inflection point. The seventh lens element has an object-side surface and an image-side surface being both aspheric.

13 Claims, 27 Drawing Sheets



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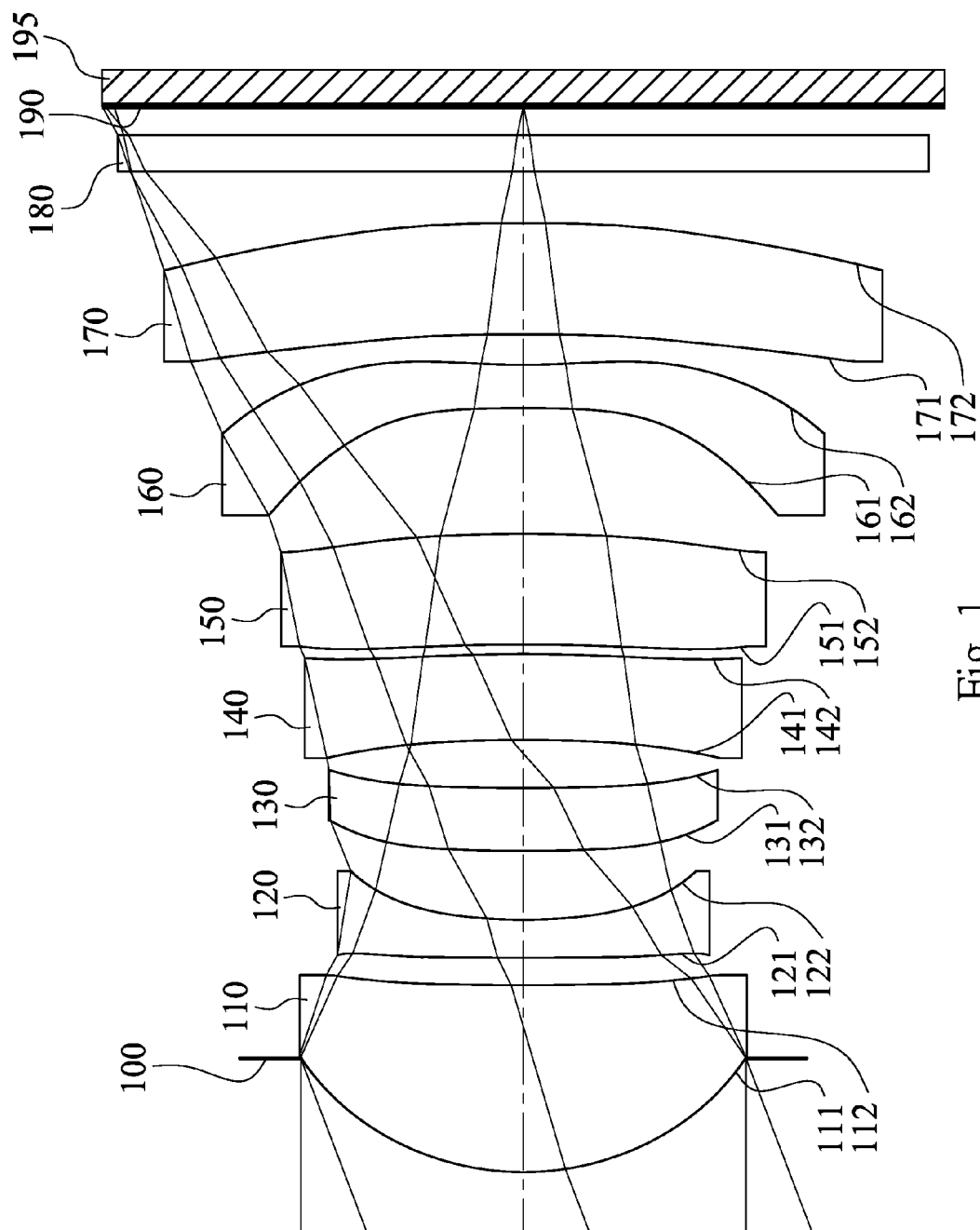


Fig. 1

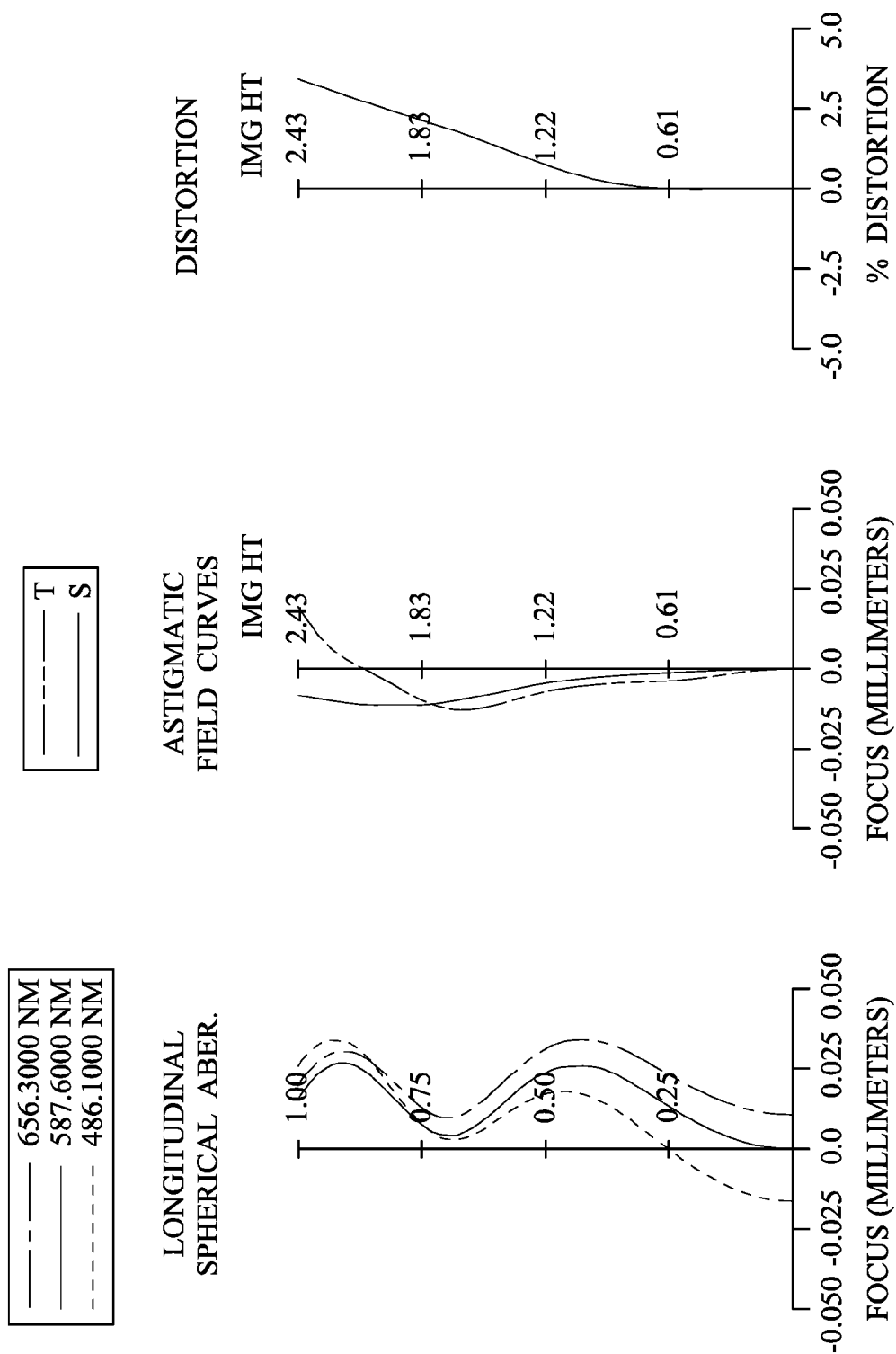


Fig. 2

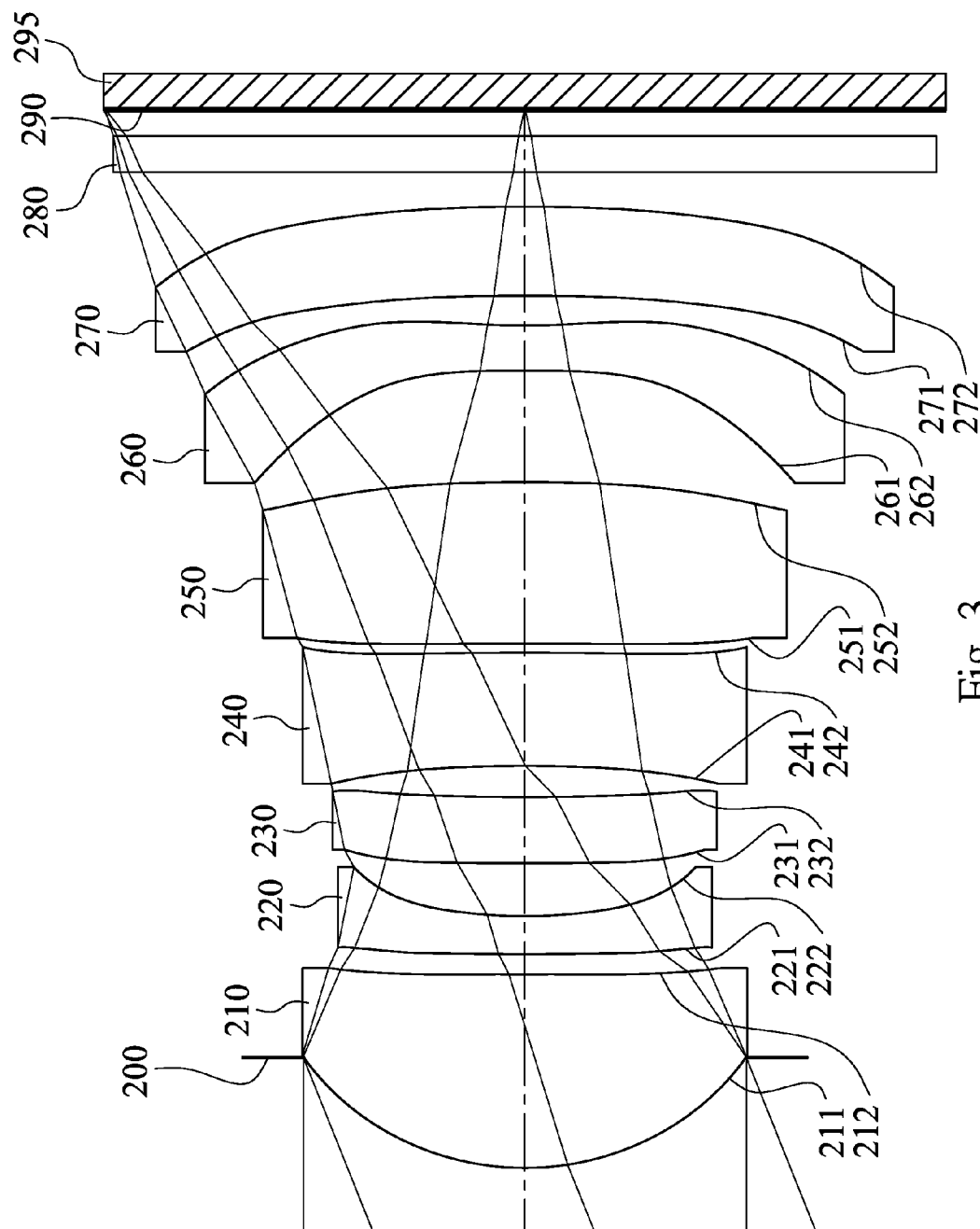


Fig. 3

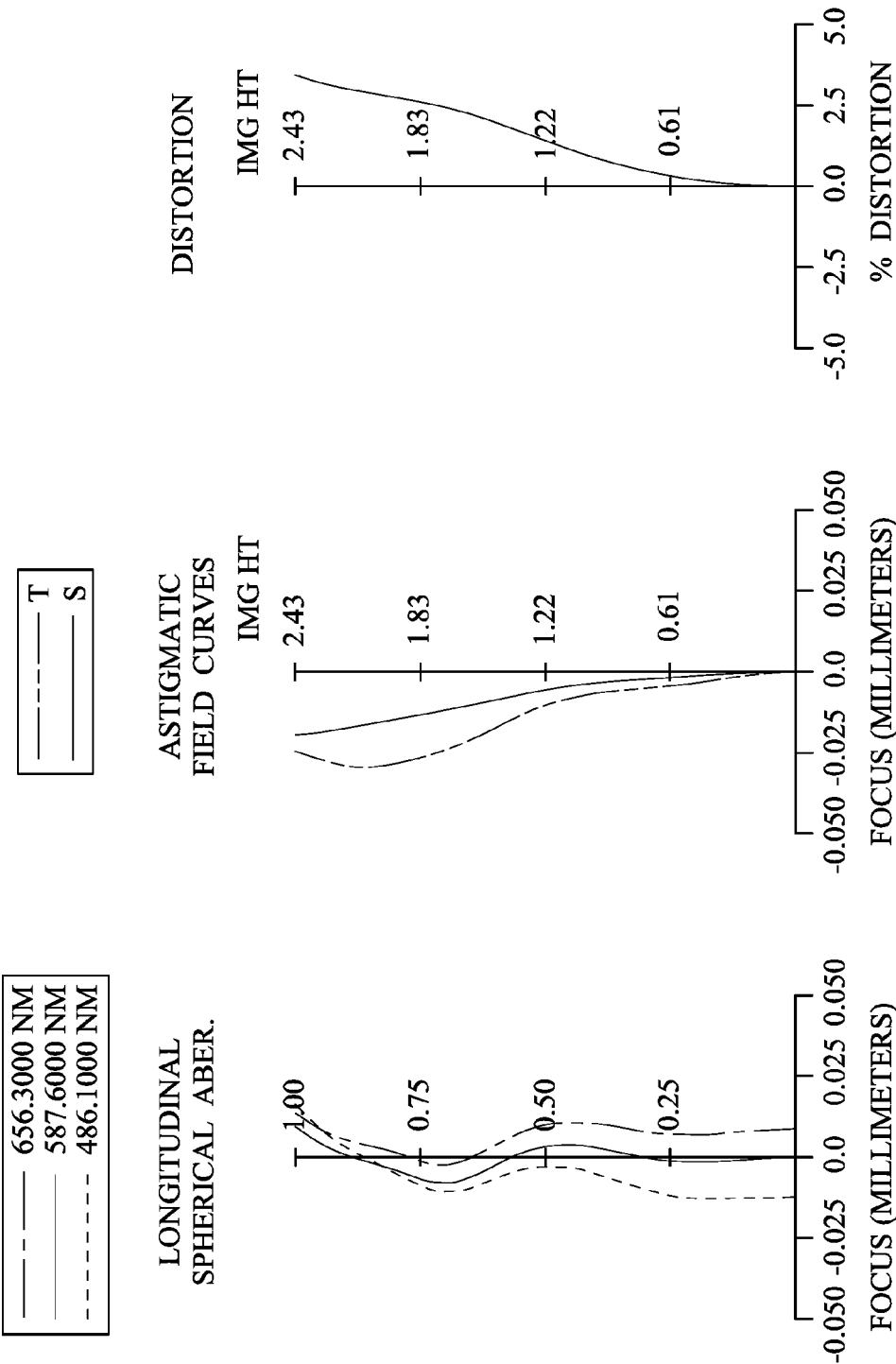


Fig. 4

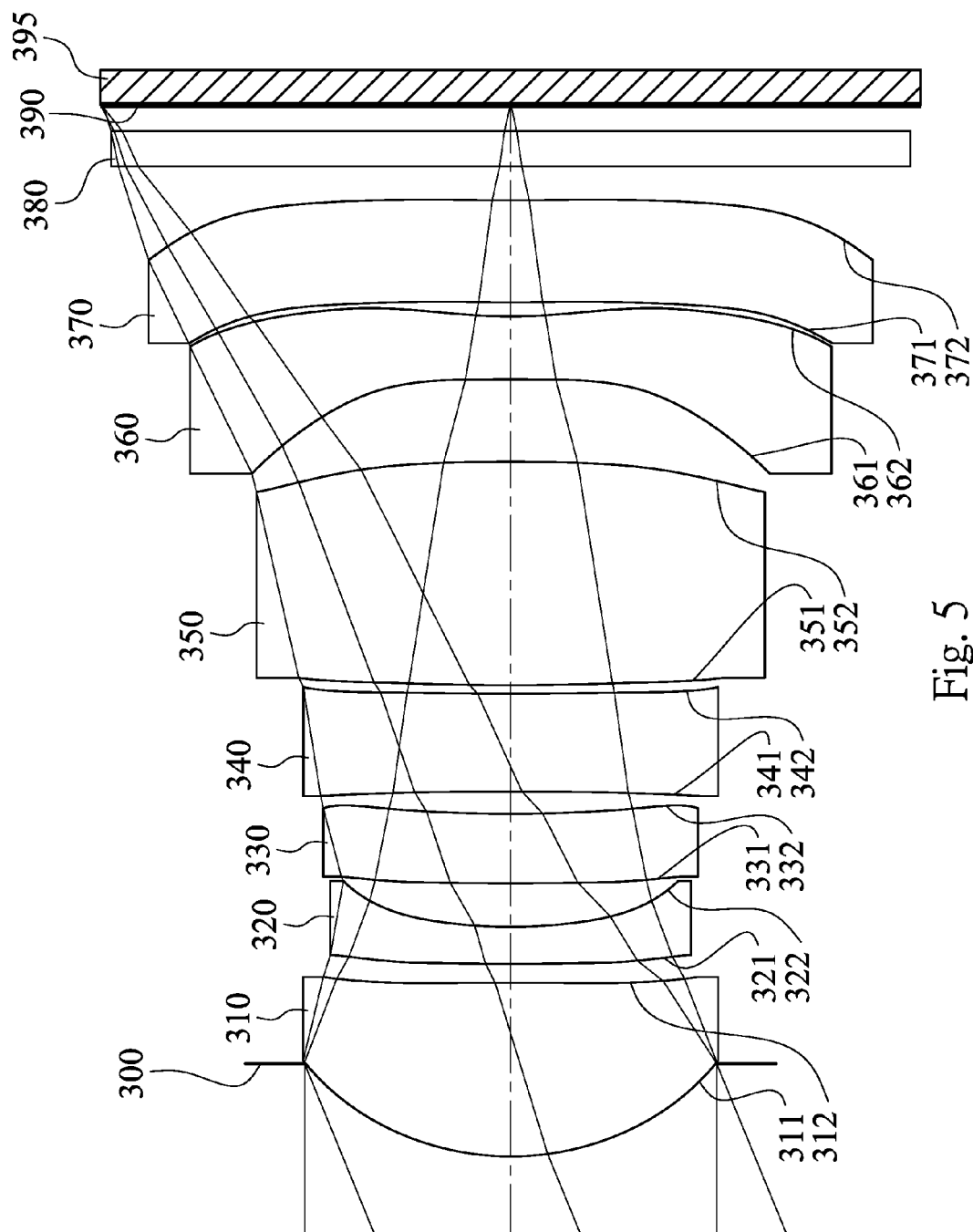


Fig. 5

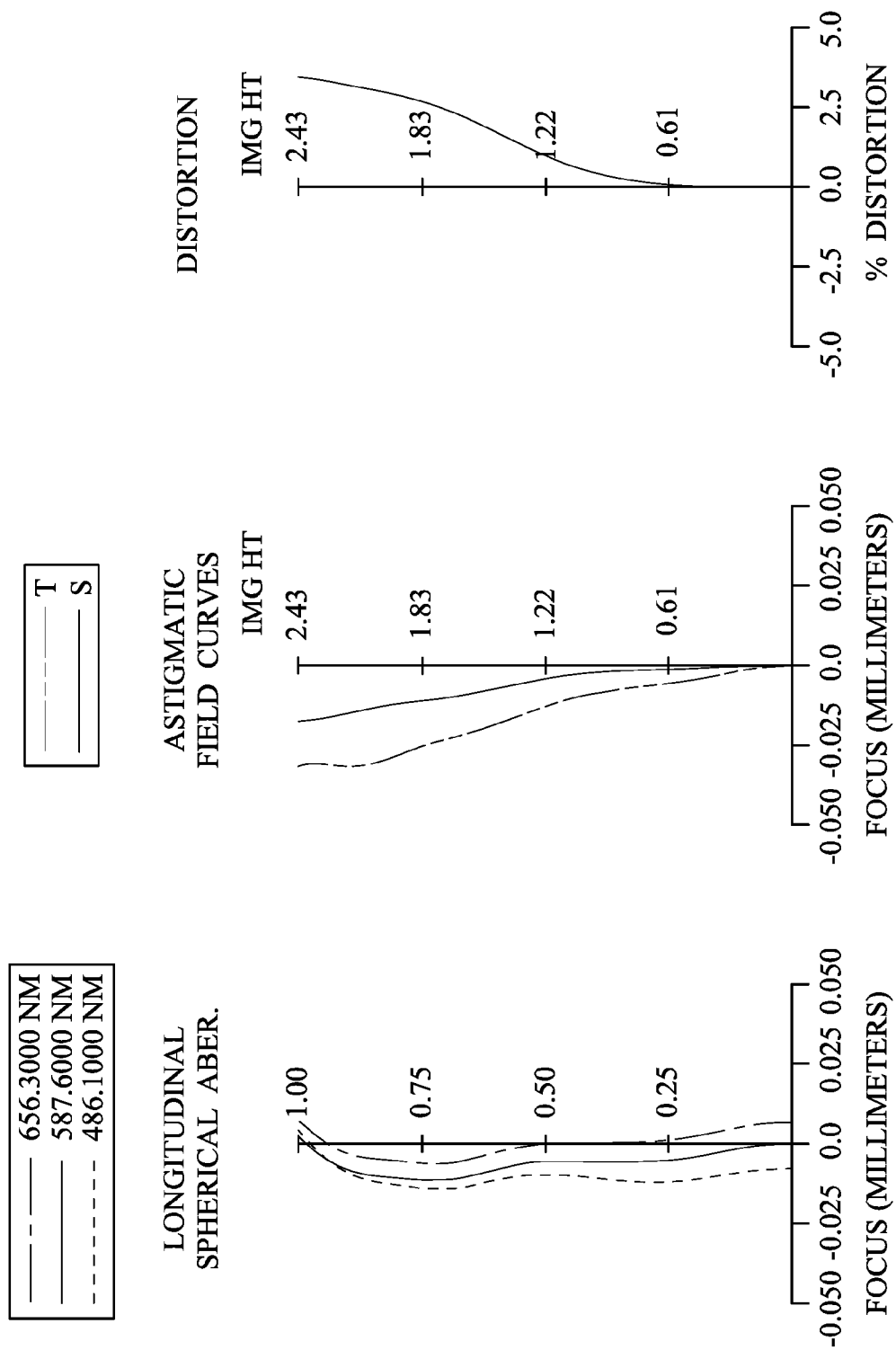


Fig. 6

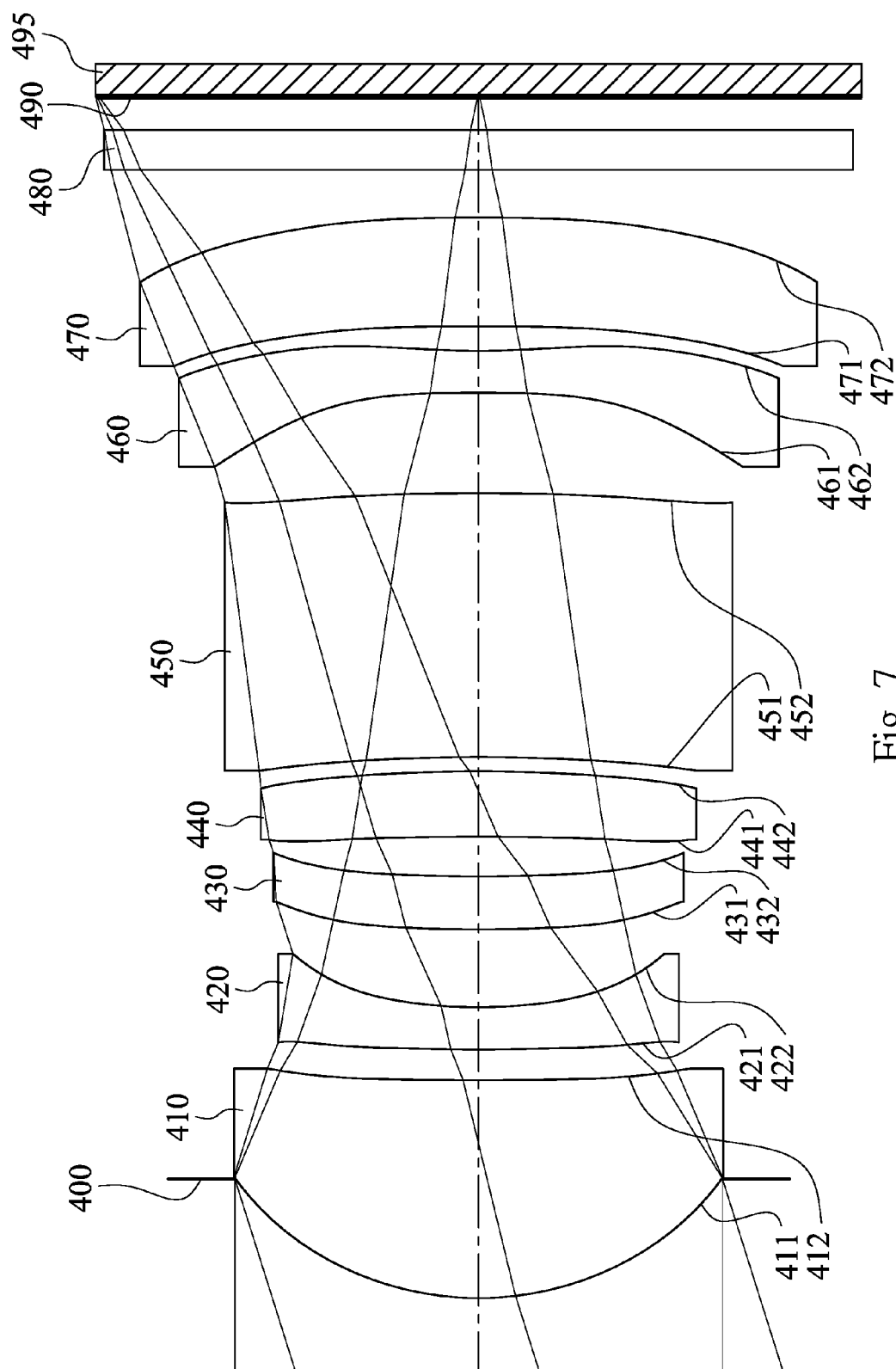
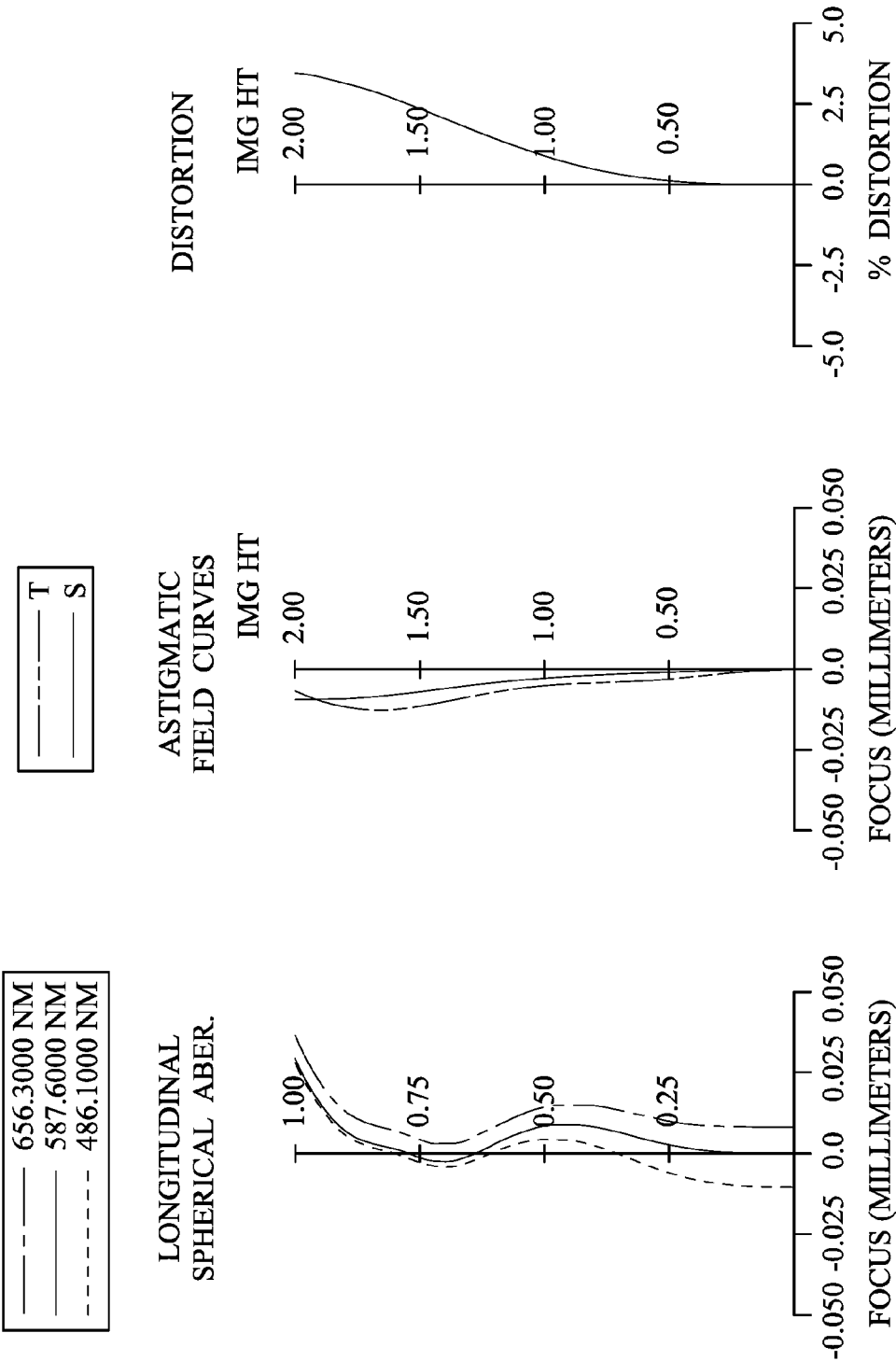


Fig. 7



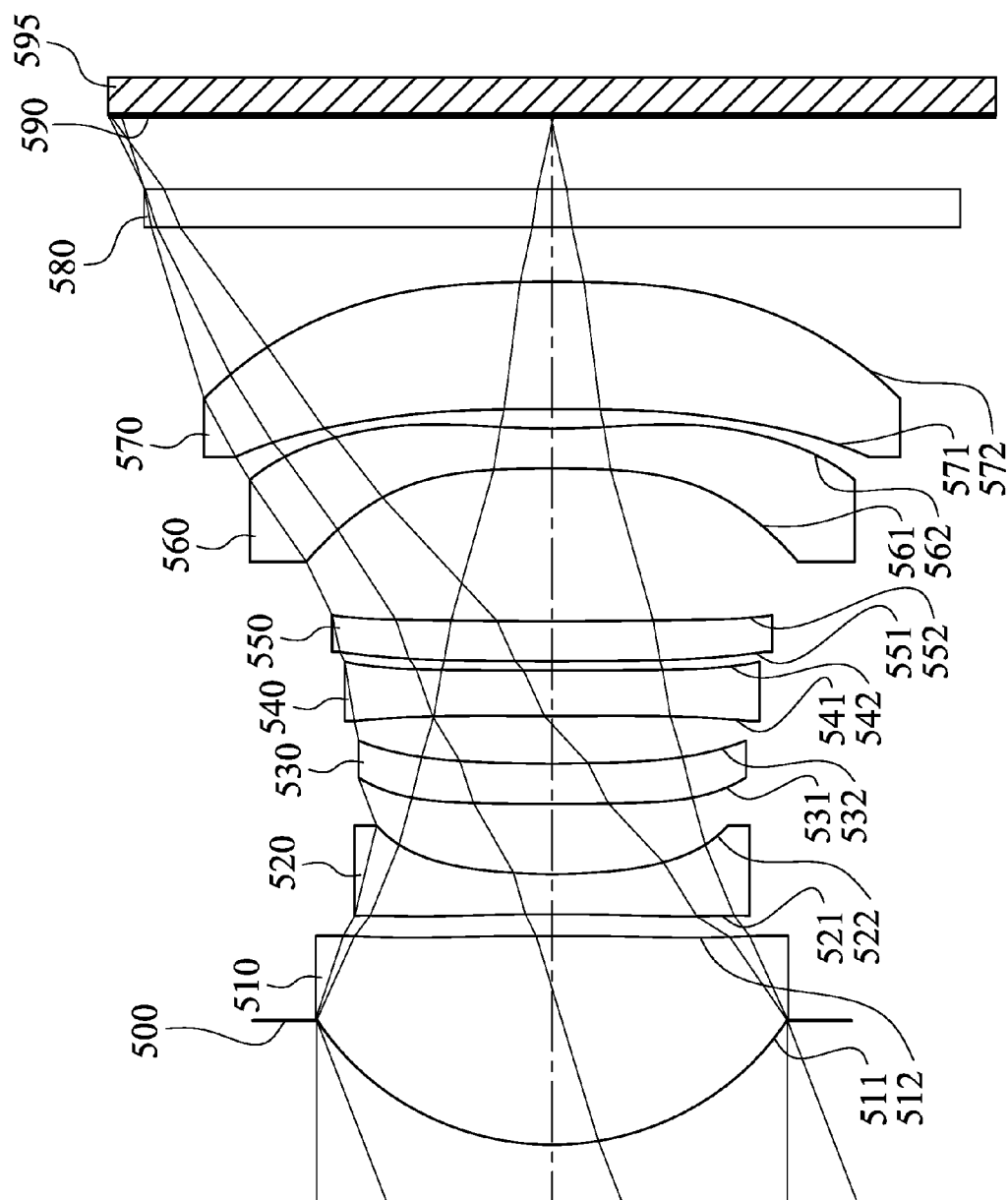


Fig. 9

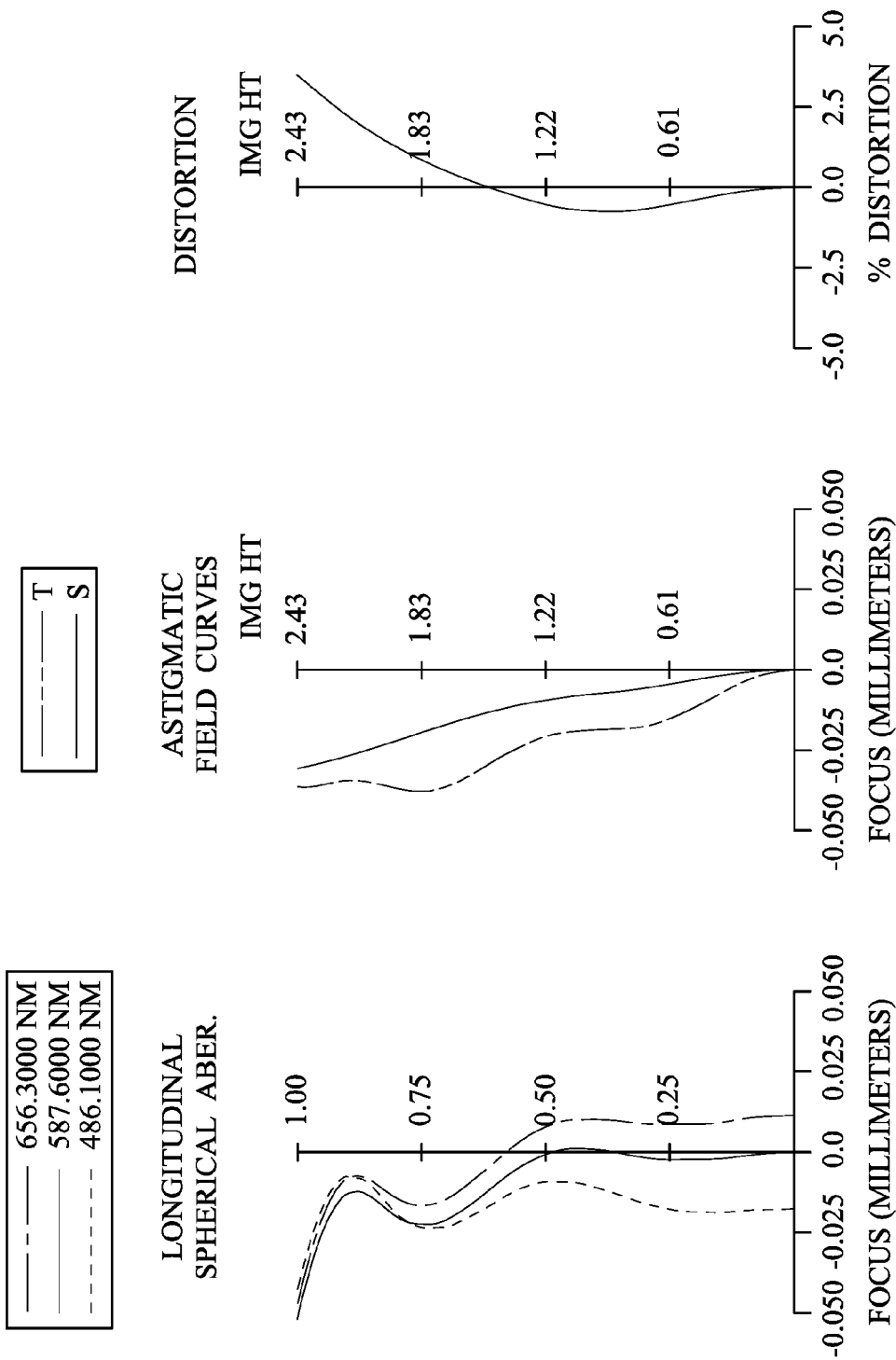


Fig. 10

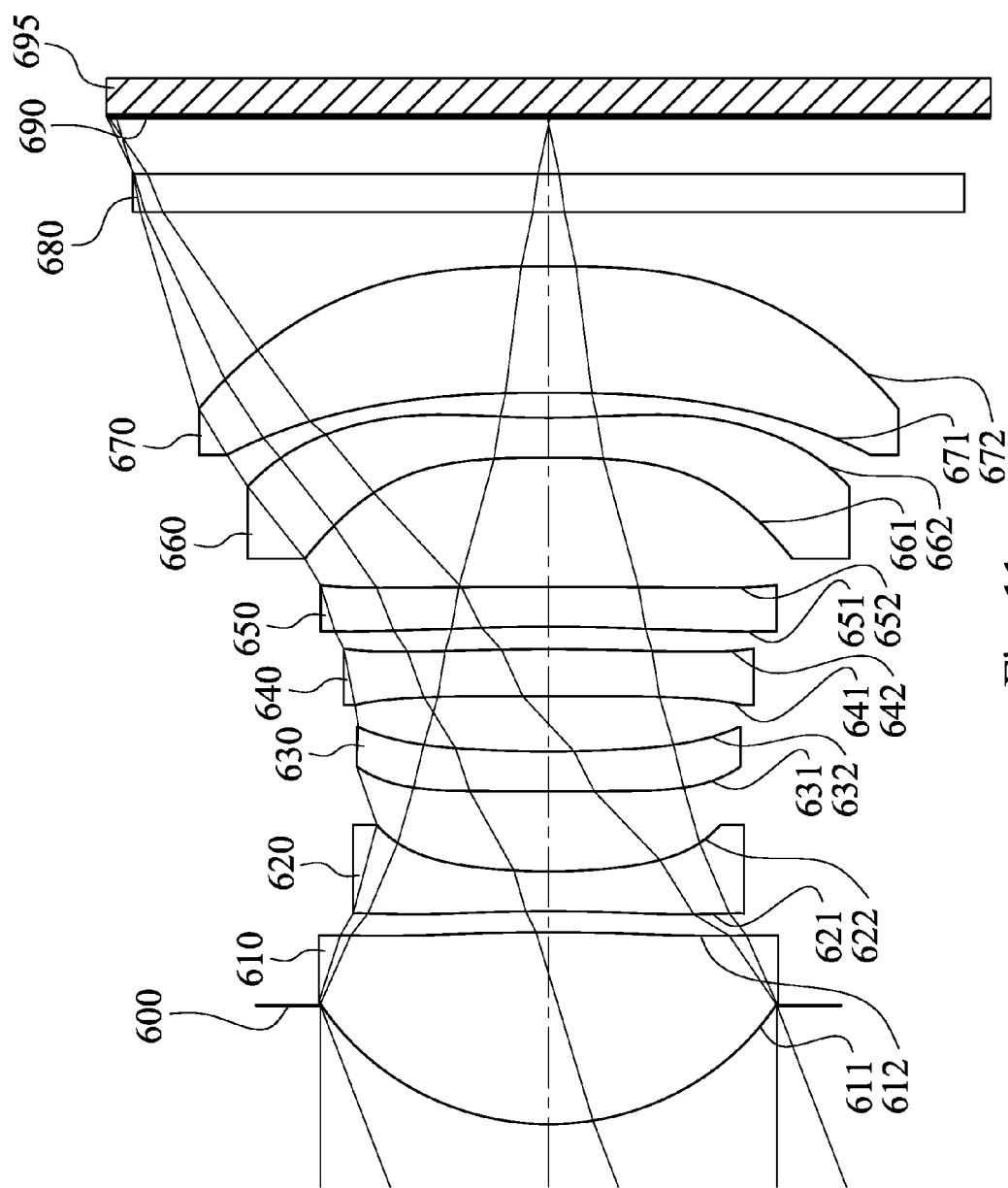


Fig. 11

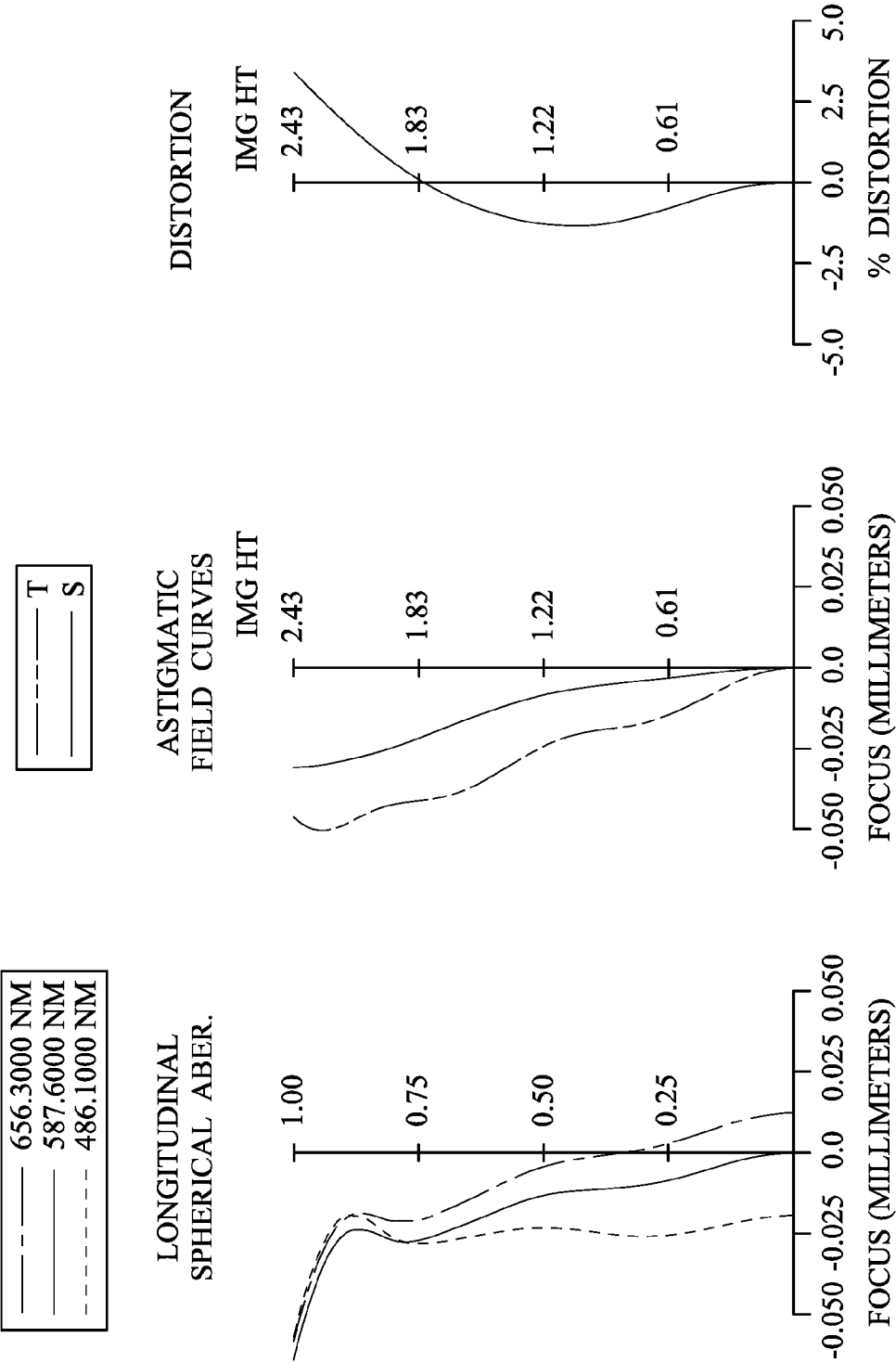
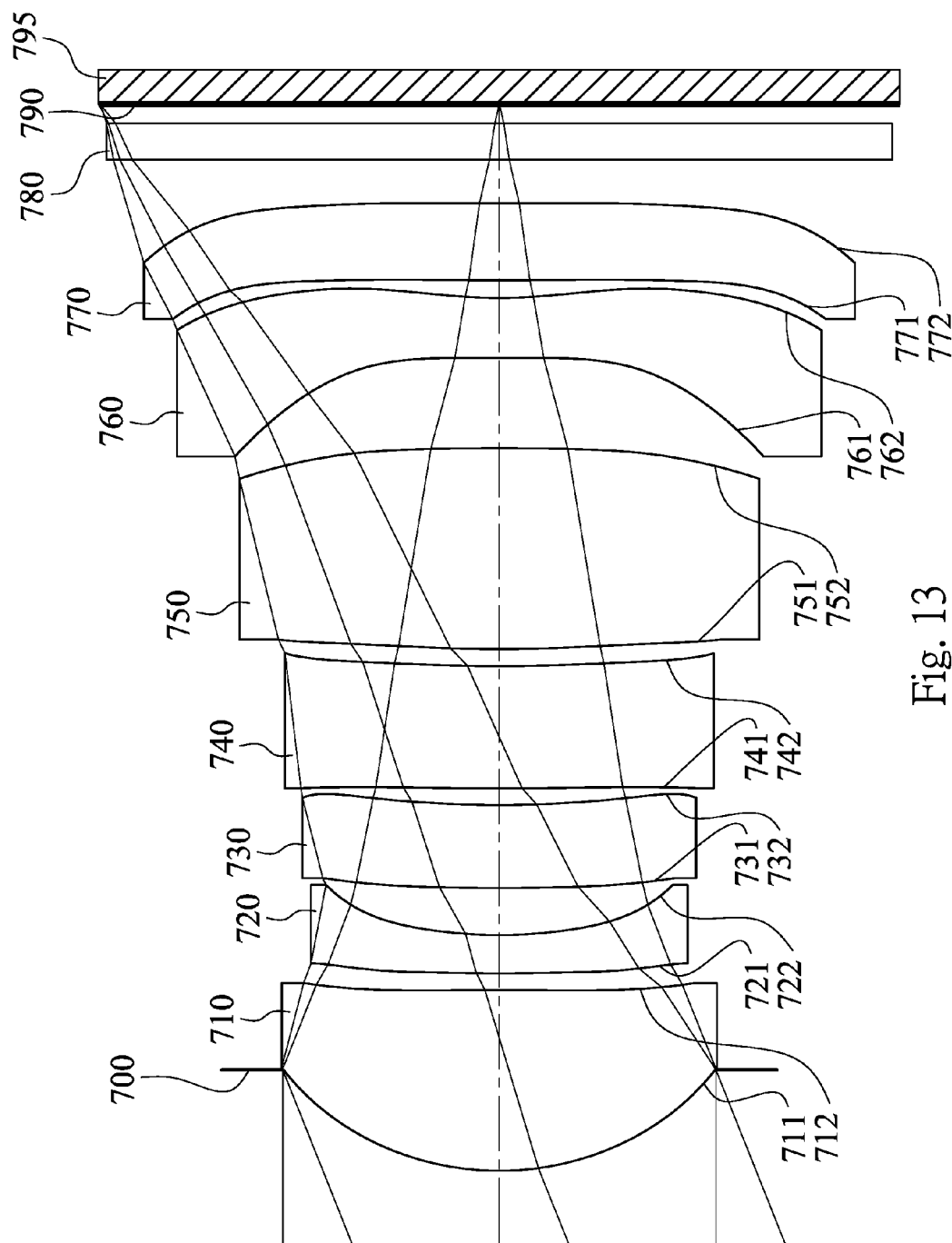


Fig. 12



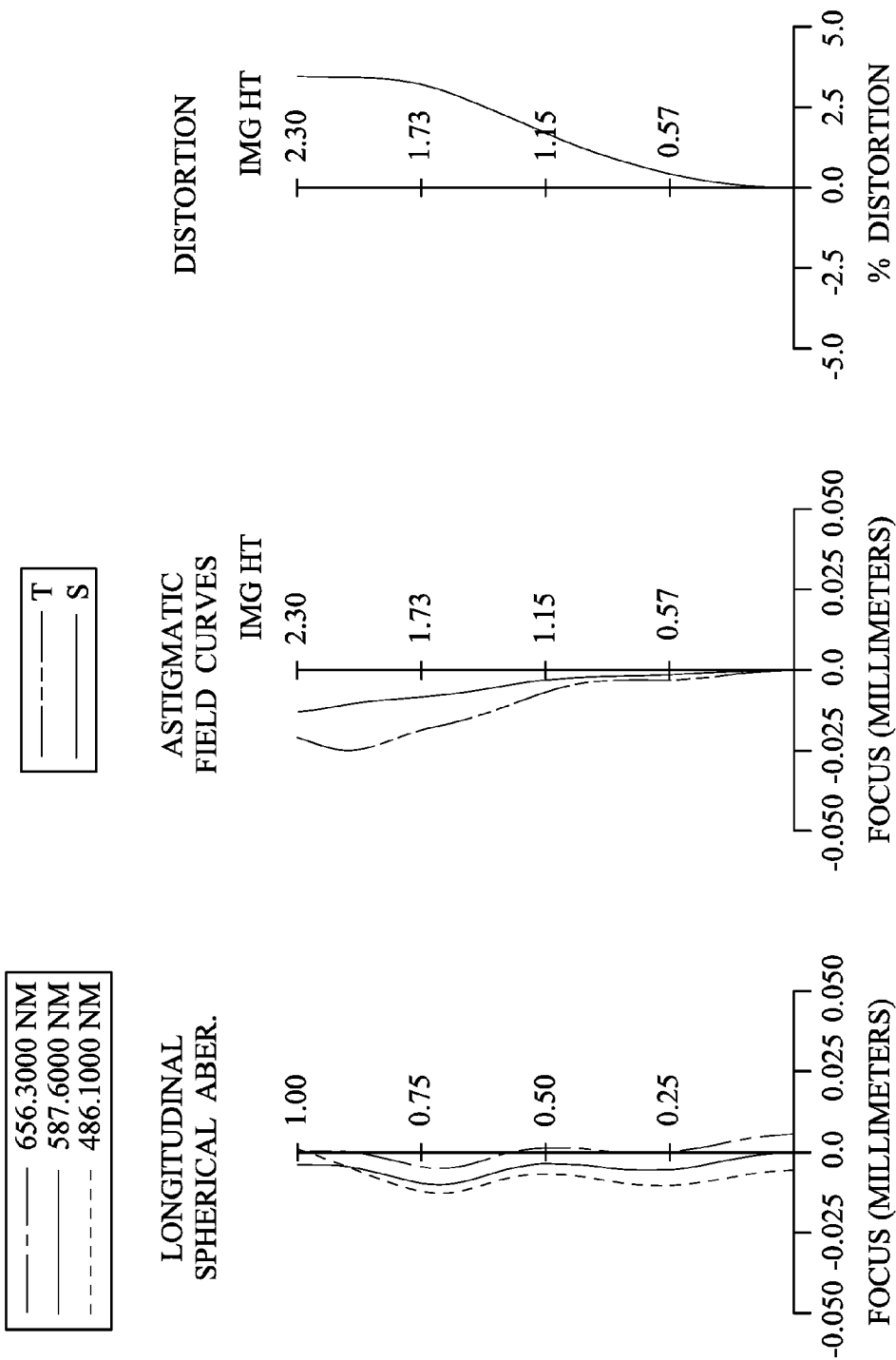


Fig. 14

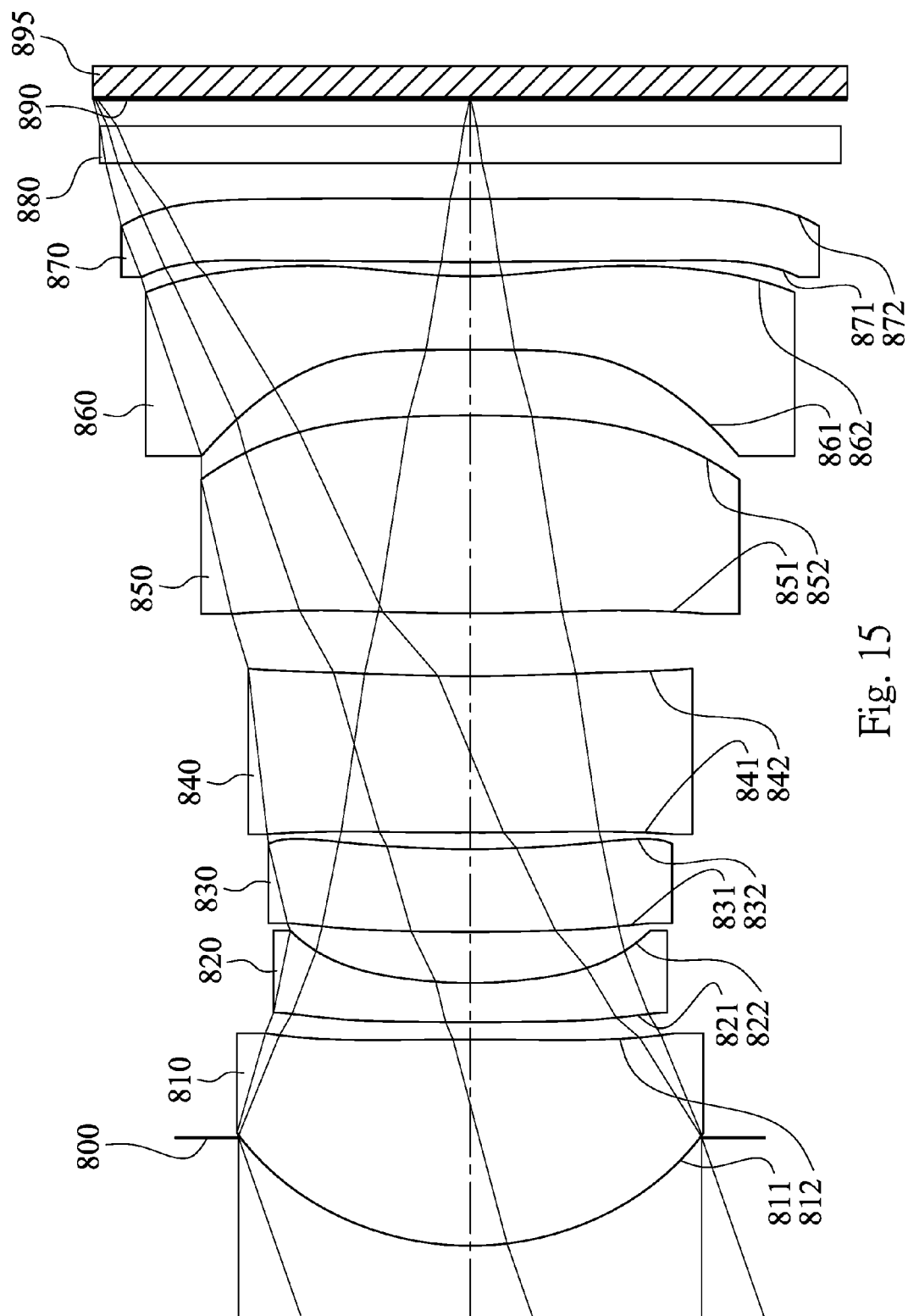
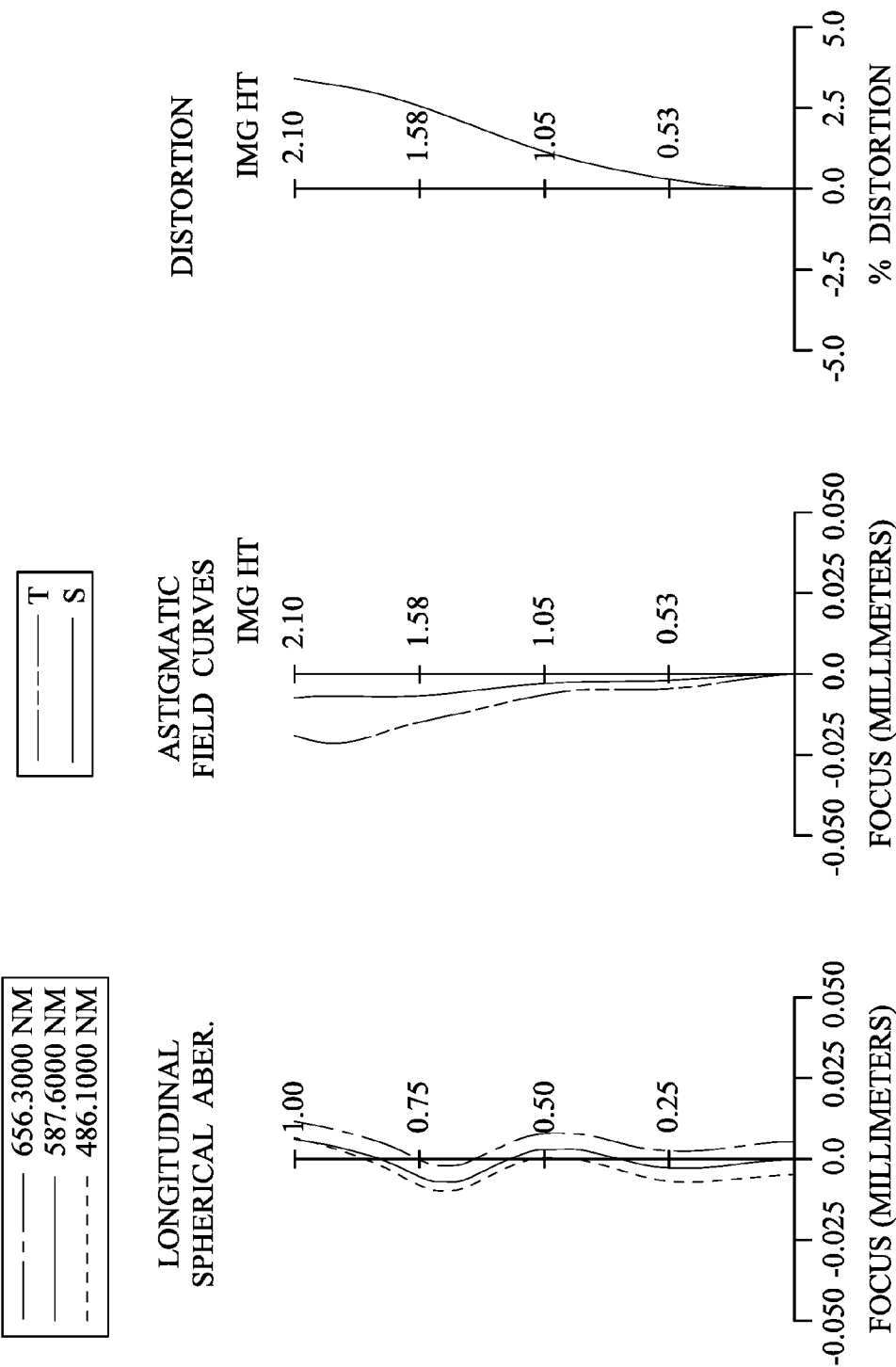


Fig. 15



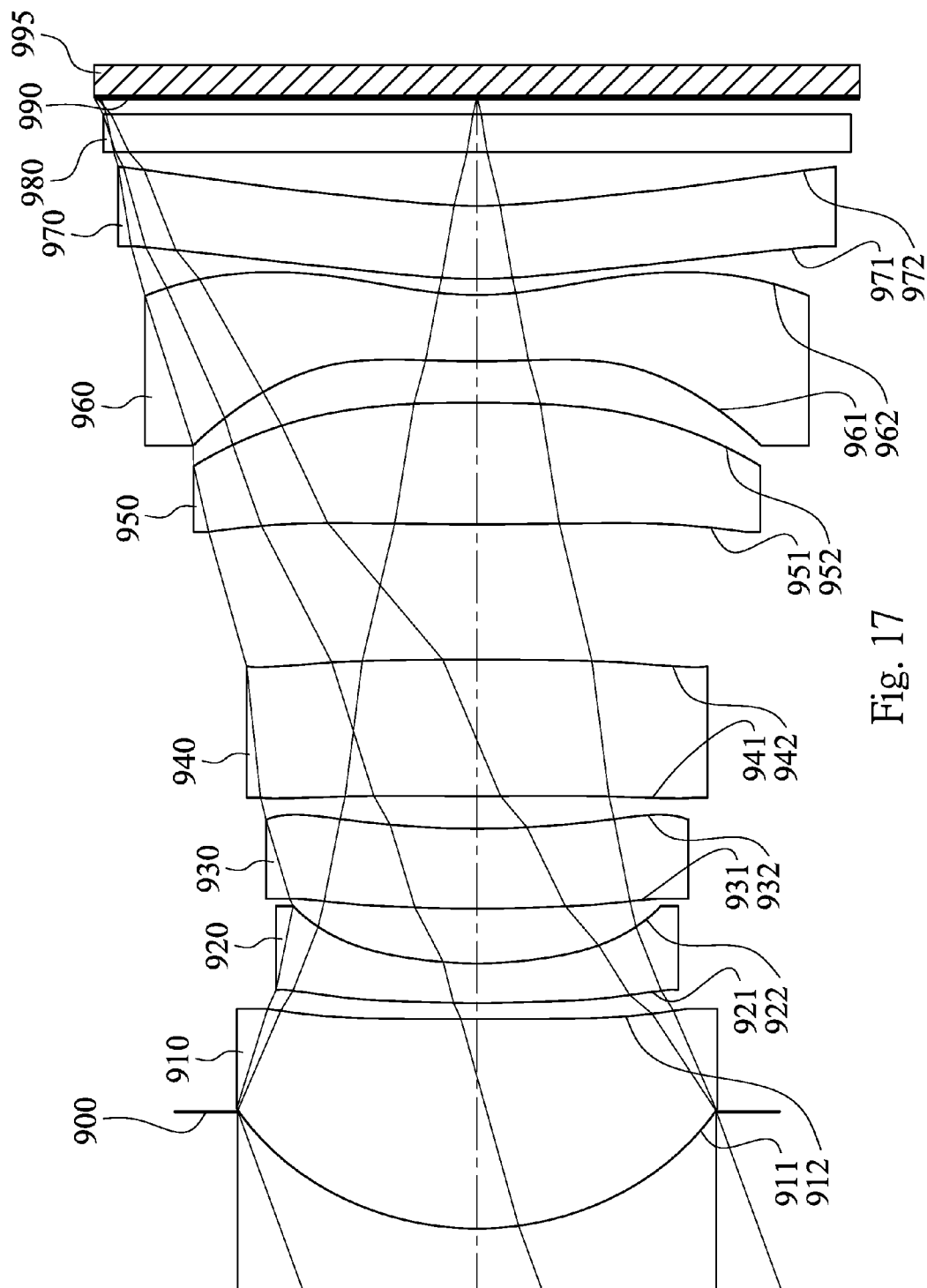


Fig. 17

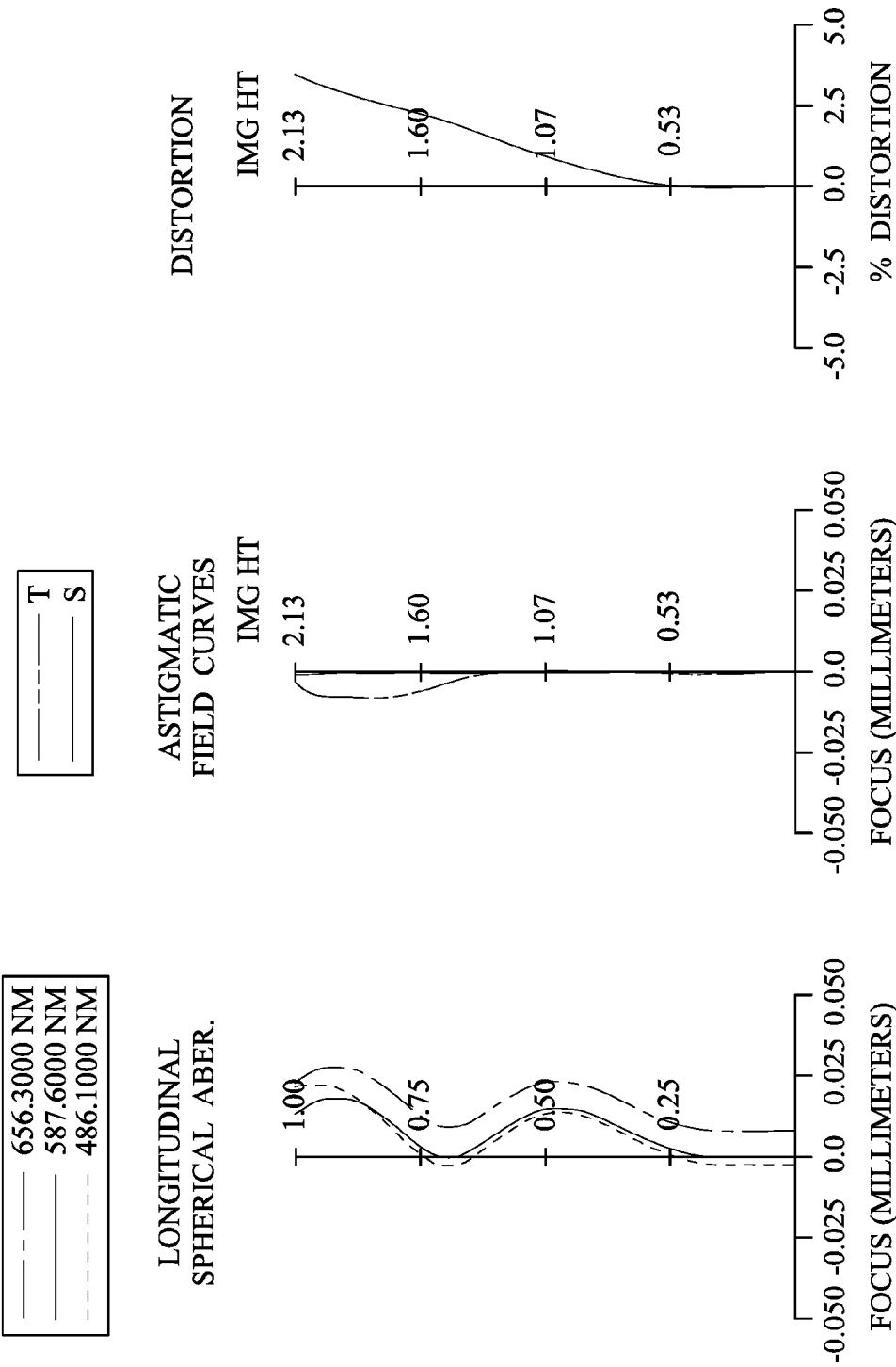


Fig. 18

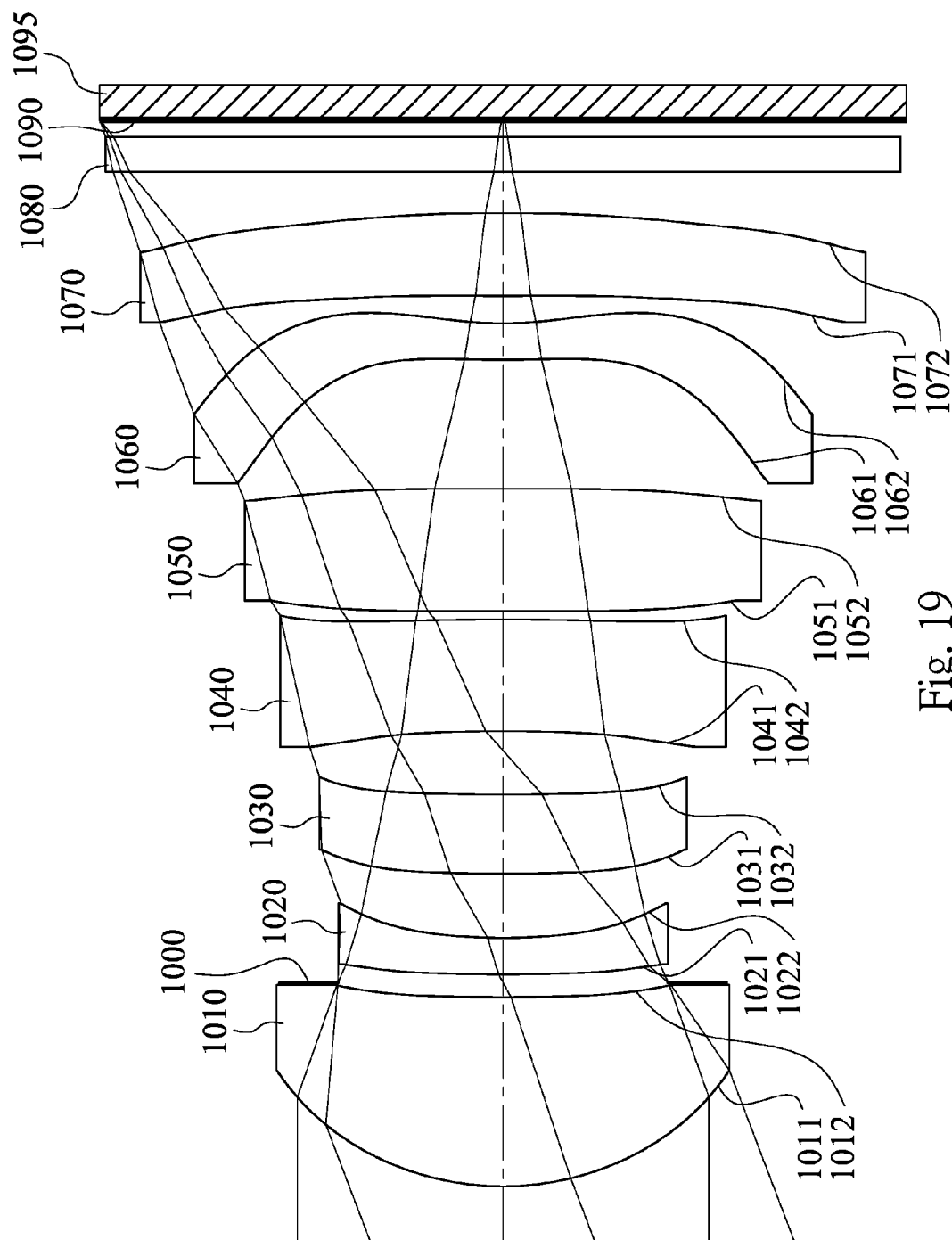
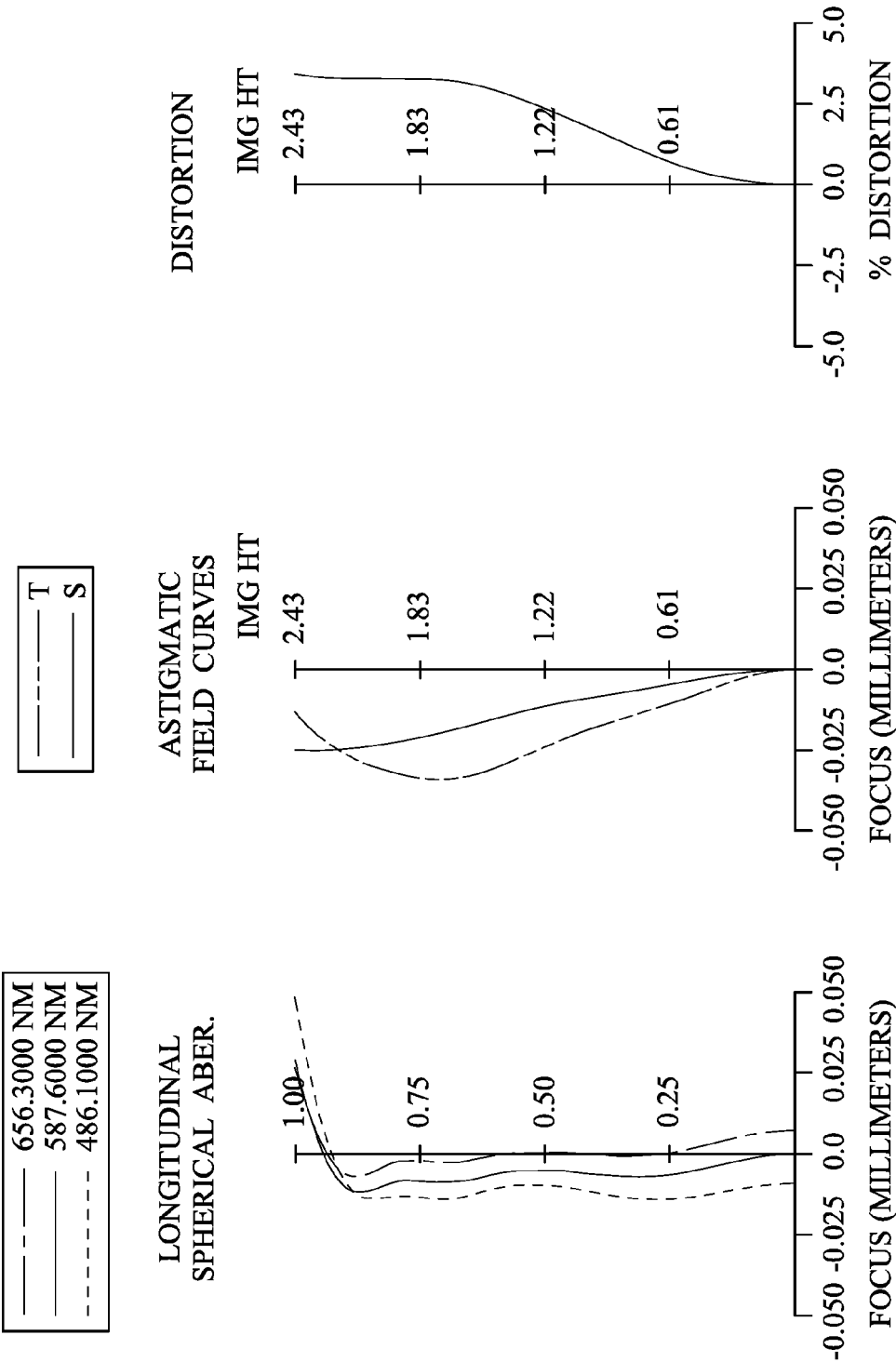


Fig. 19



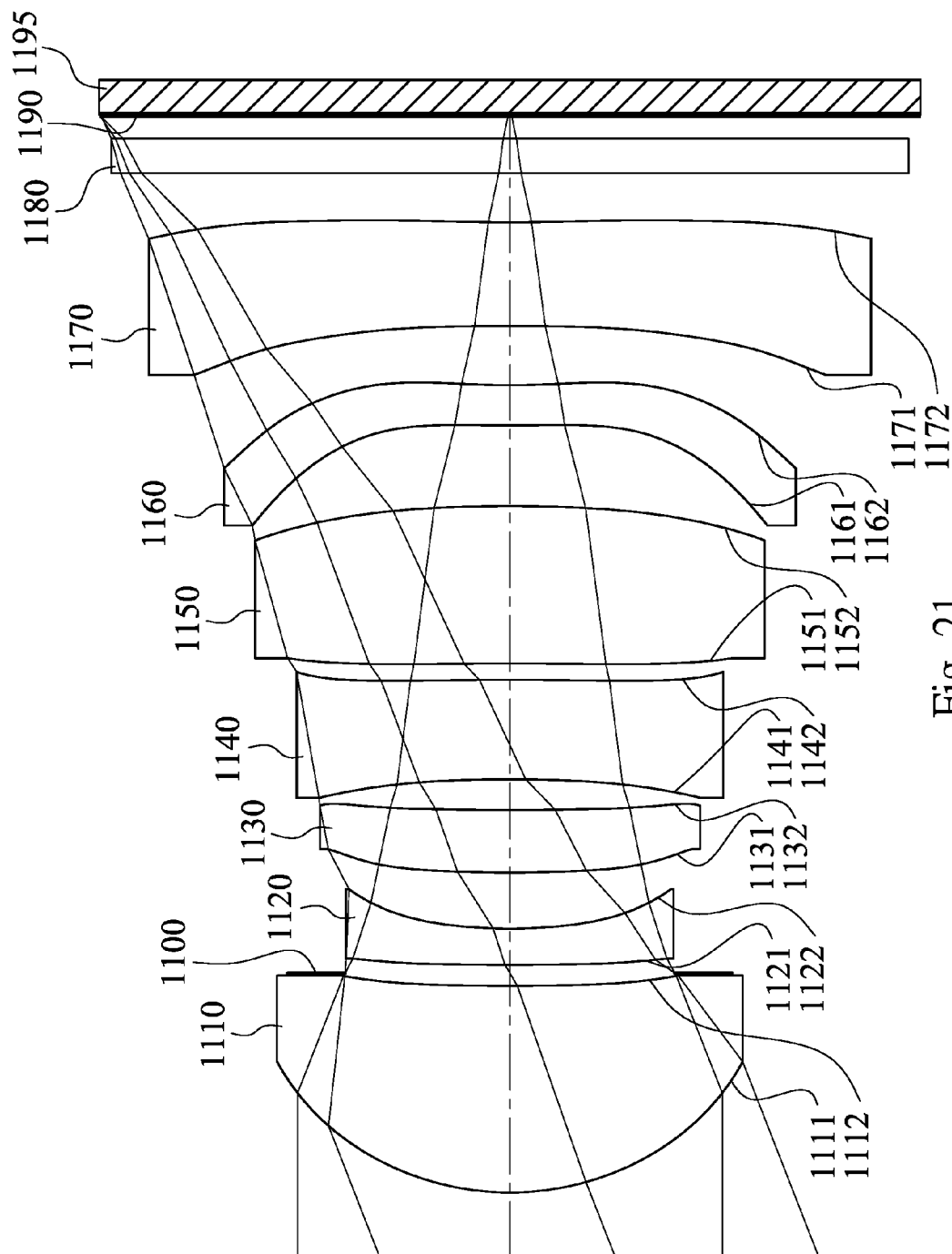


Fig. 21

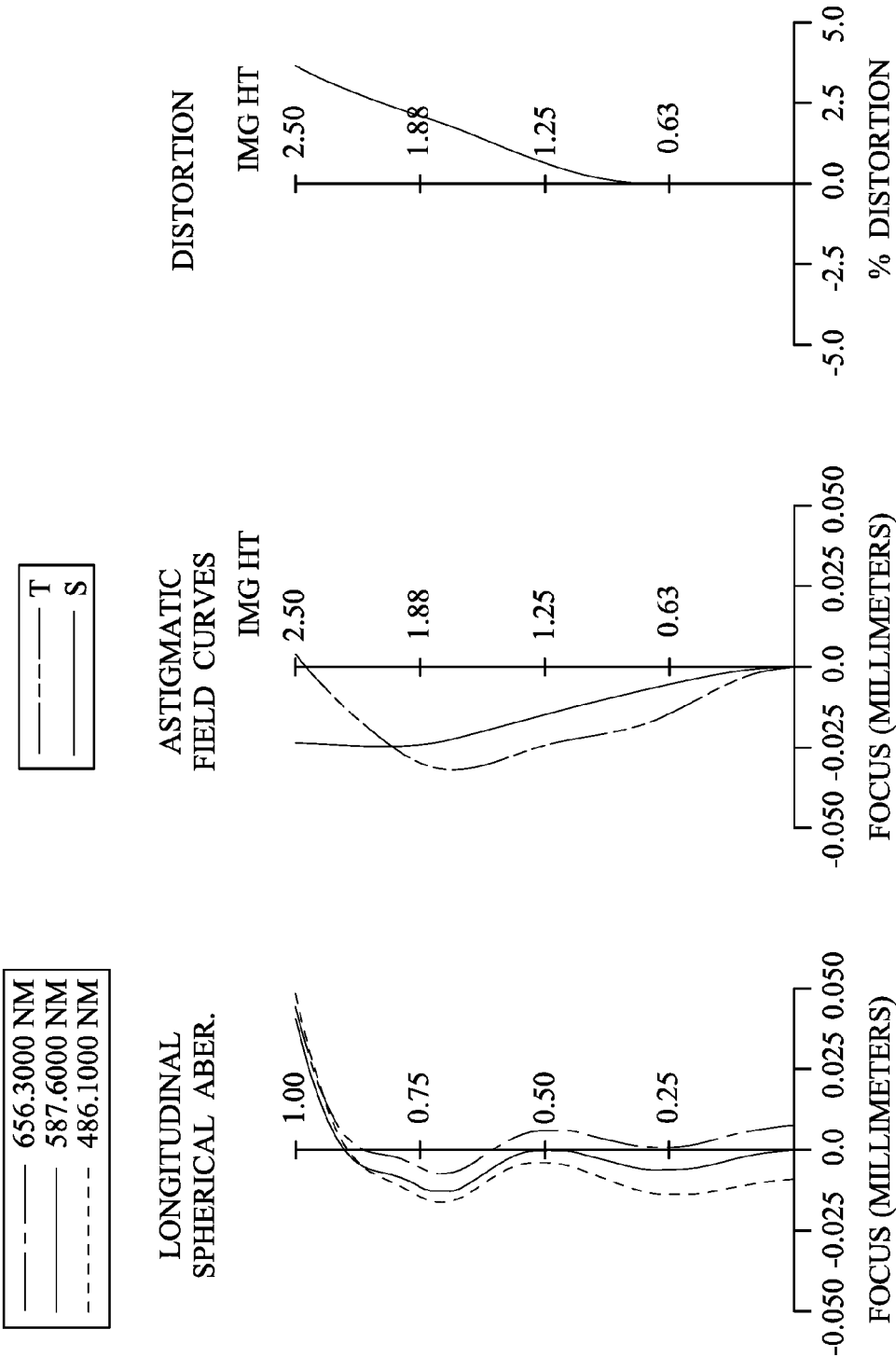


Fig. 22

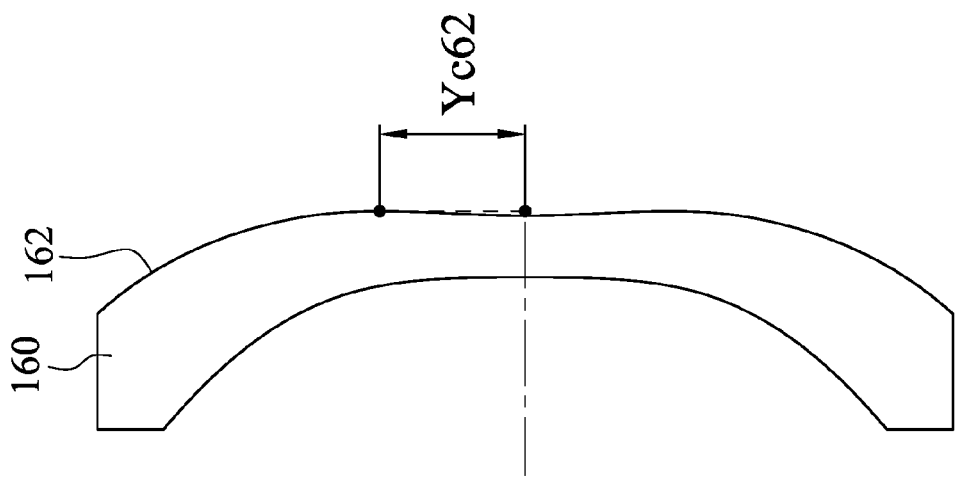


Fig. 23

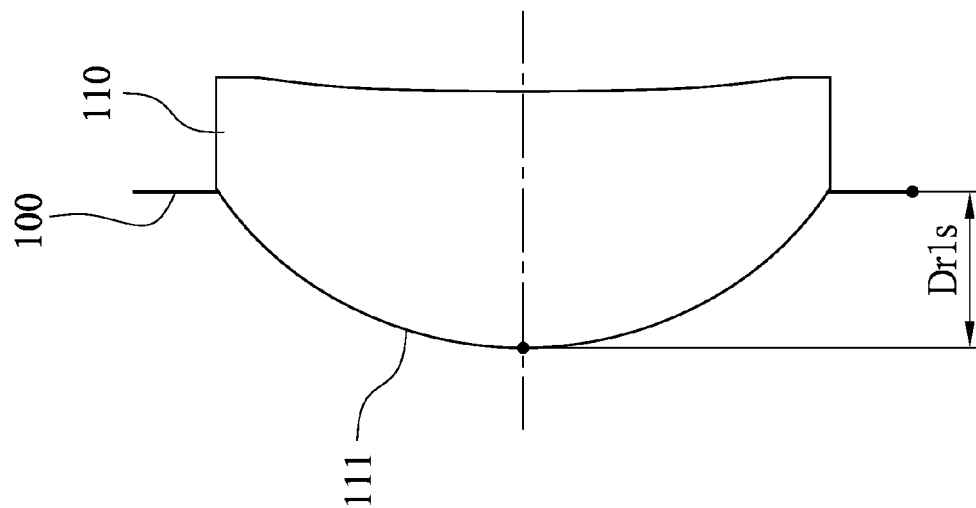


Fig. 24

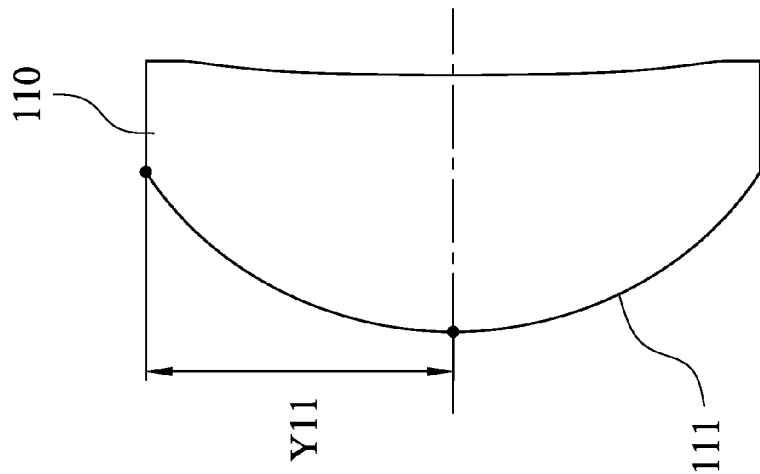


Fig. 25

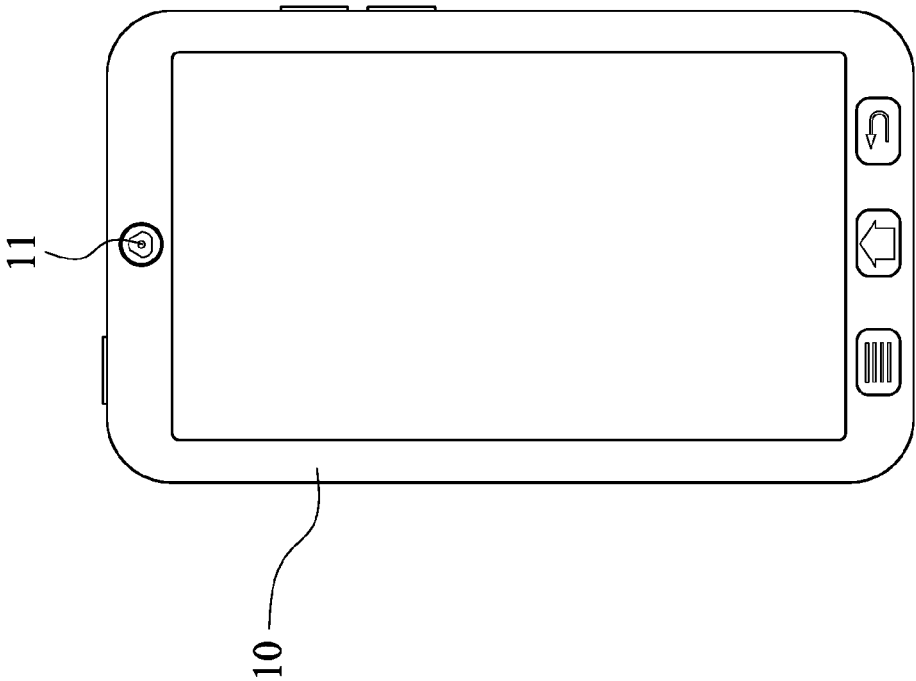


Fig. 26

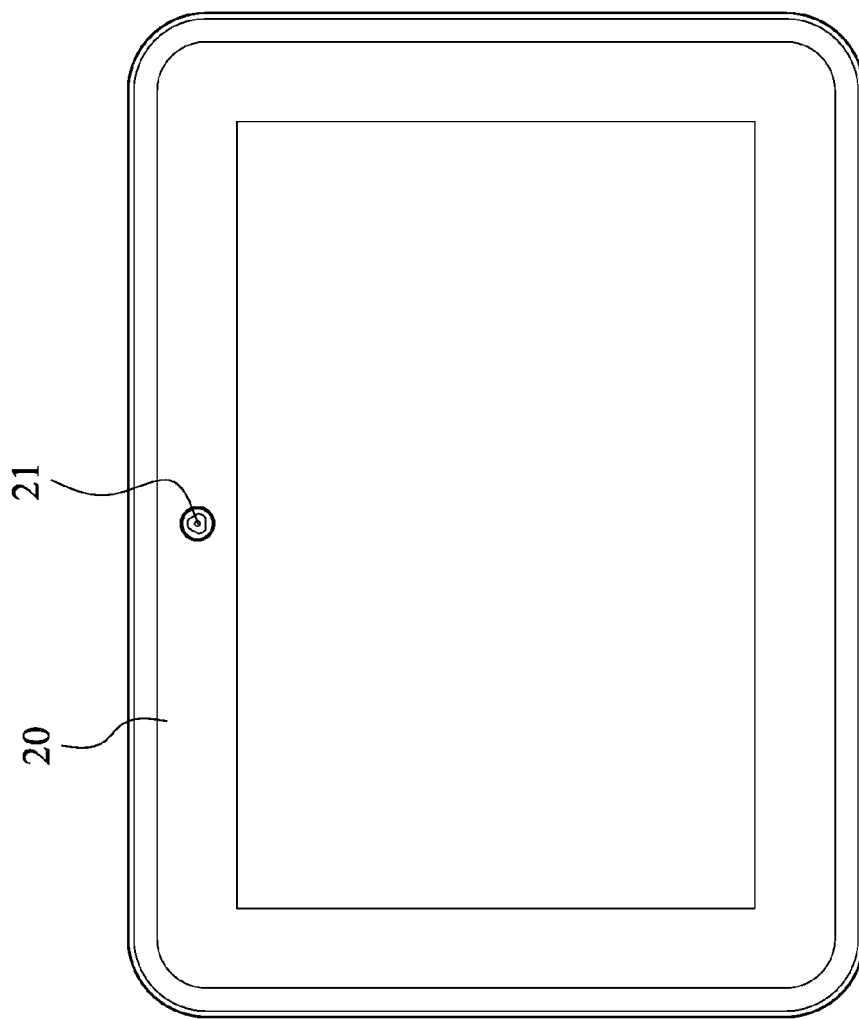


Fig. 27

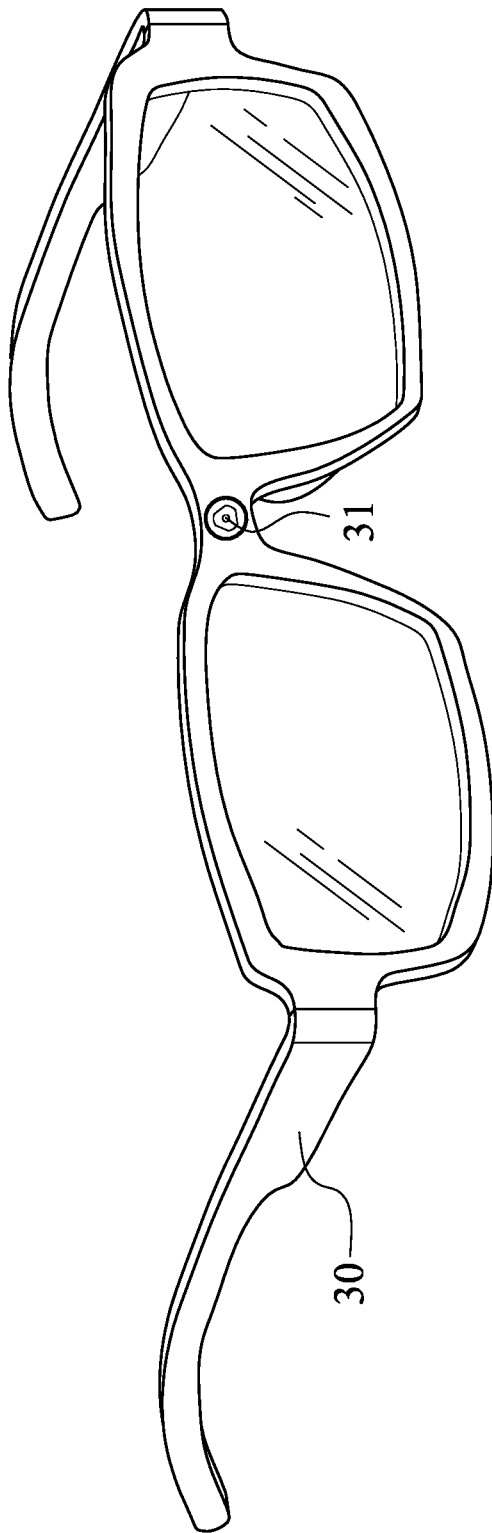


Fig. 28

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PHOTOGRAPHING OPTICAL LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND ELECTRONIC DEVICE

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 105103795, filed Feb. 4, 2016, which is herein incorporated by reference.

BACKGROUND

Technical Field

The present disclosure relates to a photographing optical lens assembly and an image capturing device. More particularly, the present disclosure relates to a photographing optical lens assembly and an image capturing device with a compact size applicable to electronic devices.

Description of Related Art

With the popularity of photographing module applications, utilizing photographing modules in various intelligent electronic devices, automobile devices, recognition devices, entertainment devices, sport devices and household intelligent assistance systems is becoming a major trend in developments of future technologies. For obtaining extensive experiences in utilizations of the photographing modules, intelligent devices with one or more lens assemblies are the market mainstream, and various lens systems with different features are developed in response to different demands.

In the conventional compact lens assemblies, the image quality has been compromised due to the common trend for system miniaturizations. While conventional imaging systems with high imaging quality usually adopt multi-element lens configuration of spherical glass lens elements, the size of the imaging systems would be too large and cumbersome. Also, the costs of the imaging systems would be too expensive for applications in various devices and products. Hence, conventional imaging systems cannot satisfy the current trend of the technological developments.

SUMMARY

According to one aspect of the present disclosure, a photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, and a seventh lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has negative refractive power. The sixth lens element has at least one of an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element comprises at least one inflection point. The seventh lens element has an object-side surface and an image-side surface being both aspheric. The photographing optical lens assembly has a total of seven lens elements. When a curvature radius of the object-side surface of the first lens element is $R1$, a focal length of the sixth lens element is $f6$, and a focal length of the seventh lens element is $f7$, the following conditions are satisfied:

$$2.85 < f/R1, \text{ and}$$

$$-2.0 < f6/f7 < 1.5.$$

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According to another aspect of the present disclosure, an image capturing device includes the photographing optical lens assembly of the aforementioned aspect and an image sensor, wherein the image sensor is disposed on an image surface of the photographing optical lens assembly.

According to yet another aspect of the present disclosure, an electronic device includes the image capturing device of the aforementioned aspect.

According to further another aspect of the present disclosure, a photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element, and a seventh lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The sixth lens element has at least one of an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element comprises at least one inflection point. The seventh lens element has an object-side surface and an image-side surface being both aspheric. The photographing optical lens assembly has a total of seven lens elements. When a curvature radius of the object-side surface of the first lens element is $R1$, a central thickness of the first lens element is $CT1$, a focal length of the sixth lens element is $f6$, and a focal length of the seventh lens element is $f7$, the following conditions are satisfied:

$$R1/CT1 < 2.5; \text{ and}$$

$$-0.90 < f6/f7 < 1.5.$$

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment;

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

FIG. 17 is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure;

FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment;

FIG. 19 is a schematic view of an image capturing device according to the 10th embodiment of the present disclosure;

FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 10th embodiment;

FIG. 21 is a schematic view of an image capturing device according to the 11th embodiment of the present disclosure;

FIG. 22 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 11th embodiment;

FIG. 23 shows a schematic view of a parameter Y_{c62} according to the 1st embodiment of FIG. 1;

FIG. 24 shows a schematic view of a parameter $Dr1s$ according to the 1st embodiment of FIG. 1;

FIG. 25 shows a schematic view of a parameter $Y11$ according to the 1st embodiment of FIG. 1;

FIG. 26 is a schematic view of an electronic device according to the 12th embodiment of the present disclosure;

FIG. 27 is a schematic view of an electronic device according to the 13th embodiment of the present disclosure; and

FIG. 28 is a schematic view of an electronic device according to the 14th embodiment of the present disclosure.

DETAILED DESCRIPTION

A photographing optical lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element, wherein the photographing optical lens assembly has a total of seven lens elements.

The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. Therefore, it is favorable for providing the main converging ability of the incident light in the photographing optical lens assembly, so that the volume of the photographing optical lens assembly can be effectively reduced to satisfy the demand of compact size.

The second lens element can have negative refractive power, and can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, focusing positions of lights with different wavelengths can be balanced so as to avoid the image overlay, and the spherical aberration can be reduced by a coordinating configuration with the first lens element.

The fourth lens element can have an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. Therefore, it is favorable for correcting the astigmatism of the photographing optical lens assembly by balancing the light path in the tangential direction and the sagittal direction. Furthermore, at least one of the object-side surface and the image-

side surface of the fourth lens element includes at least one inflection point, so that aberrations in an off-axial region thereof can be corrected.

The fifth lens element can have an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. Therefore, it is favorable for correcting aberrations of the photographing optical lens assembly.

The sixth lens element can have negative refractive power and can have an image-side surface being concave in a paraxial region thereof. Therefore, the Petzval Surface can be flatter, and the principal point can be positioned closer to the object side so as to reduce the back focal length for controlling the total track length. Furthermore, at least one of the object-side surface and the image-side surface of the sixth lens element includes at least one inflection point. Hence, it is favorable for correcting aberrations in the off-axial region and reducing the back focal length so as to balance between the image quality and the physical size of the lens assembly.

The seventh lens element can have positive refractive power and can have an image-side surface being convex in a paraxial region thereof. Therefore, the symmetrical structure of the photographing optical lens assembly can be formed by the seventh lens element and the first lens element for improving the image quality.

When a focal length of the photographing optical lens assembly is f , and a curvature radius of the object-side surface of the first lens element is $R1$, the following condition is satisfied: $2.85 < f/R1$. Therefore, it is favorable for obtaining the telephoto characteristic and controlling the total track length of the photographing optical lens assembly at the same time so as to obtain the demand of compact size thereof. Preferably, the following condition can be satisfied: $3.10 < f/R1 < 7.50$.

When a focal length of the sixth lens element is $f6$, and a focal length of the seventh lens element is $f7$, the following conditions are satisfied: $-2.0 < f6/f7 < 1.5$. Therefore, the sixth lens element can have sufficiently more refractive power than the seventh lens element, so that aberrations generated from the sixth lens element can be corrected by the seventh lens element. Preferably, the following condition can be satisfied: $-0.90 < f6/f7 < 1.5$. More preferably, the following condition can be satisfied: $-0.60 < f6/f7 < 0.60$. Furthermore preferably, the following condition can be satisfied: $-0.40 < f6/f7 < 0.40$.

When the curvature radius of the object-side surface of the first lens element is $R1$, and a central thickness of the first lens element is $CT1$, the following condition is satisfied: $R1/CT1 < 2.5$. Therefore, the first lens element can have sufficient positive refractive power for providing the better telephoto feature of the photographing optical lens assembly. Preferably, the following condition can be satisfied: $R1/CT1 < 2.2$. More preferably, the following condition can be satisfied: $R1/CT1 < 1.8$.

When the focal length of the photographing optical lens assembly is f , and a maximal image height of the photographing optical lens assembly is $ImgH$, the following condition is satisfied: $2.20 < f/ImgH < 5.50$. Therefore, it is favorable for controlling the imaging field and the angle of view effectively to enhance the resolution of the specified image area for better telephoto effects.

When the focal length of the photographing optical lens assembly is f , and a curvature radius of the image-side surface of the seventh lens element is $R14$, the following condition is satisfied: $f/R14 < 1.0$. Therefore, it is favorable for obtaining the telephoto feature and enhancing the sym-

metry of the entire photographing optical lens assembly by effectively controlling the surface shape of the lens element closest to the image surface.

When a refractive power of the first lens element is P1, a refractive power of the second lens element is P2, a refractive power of the third lens element is P3, a refractive power of the fourth lens element is P4, a refractive power of the fifth lens element is P5, a refractive power of the sixth lens element is P6, and a refractive power of the seventh lens element is P7, the following condition is satisfied: $(P3+P4+P5+P7)/(P1+P2+P6)<0.50$. Therefore, the refractive power arrangement of the photographing optical lens assembly can be balanced between the object side and the image side, and the symmetry of the entire lens assembly can be enhanced so as to reduce the sensitivity.

When an Abbe number of the seventh lens element is V7, the following condition is satisfied: $V7<30$. Therefore, it is favorable for balancing the chromatic aberration of the photographing optical lens assembly.

The photographing optical lens assembly can further include an aperture stop, which can be disposed between an imaged object and the third lens element, or further disposed between the imaged object and the first lens element.

When an axial distance between the object-side surface of the first lens element and the aperture stop is Dr1s, and a central thickness of the second lens element is CT2, the following condition is satisfied: $2.0<|Dr1s|/CT2<5.0$. Therefore, it is favorable for controlling the size of the photographing optical lens assembly by effectively allocating the aperture stop, and for enhancing the manufacturability of lens molding by controlling lens thicknesses.

When a maximal optical effective radius of the object-side surface of the first lens element is Y11, and the maximal image height of the photographing optical lens assembly is ImgH, the following condition is satisfied: $0.45<Y11/ImgH<1.0$. Therefore, it is favorable for obtaining the sufficient light by balancing the light incident range and the imaging area, so that the image illumination can be enhanced.

When a vertical distance between a non-axial critical point on the image-side surface of the sixth lens element and an optical axis is Yc62, and a central thickness of the sixth lens element is CT6, the following condition is satisfied: $0.5<Yc62/CT6<7.5$. Therefore, it is favorable for correcting aberrations in an off-axial region, and controlling the curvature of the image field effectively.

When an axial distance between the image-side surface of the seventh lens element and an image surface is BL, and the maximal image height of the photographing optical lens assembly is ImgH, the following condition is satisfied: $0.10<BL/ImgH<0.40$. Therefore, the back focal length can be controlled so as to minimize the volume of the photographing optical lens assembly for obtaining the compact size thereof.

When the focal length of the photographing optical lens assembly is f, and a focal length of the fourth lens element is f4, the following condition is satisfied: $|f/f4|<0.35$. Therefore, it is favorable for the fourth lens element to obtain the aberrations correcting ability.

When an axial distance between the fifth lens element and the sixth lens element is T56, and a sum of axial distances between every two of the lens elements of the photographing optical lens assembly that are adjacent to each other is ΣAT , the following condition is satisfied: $0.40<T56/(\Sigma AT-T56)$. Therefore, it is favorable for obtaining the telephoto characteristic by controlling the light path between the fifth lens element and the sixth lens element.

Among the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element, the sixth lens element and the seventh lens element, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. Therefore, it is favorable for obtaining various photographing ranges by effectively controlling the arrangement of the light dispersion ability.

When an axial distance between the object-side surface of the first lens element and an image surface is TL, and the focal length of the photographing optical lens assembly is f, the following condition is satisfied: $0.70<TL/f\leq 1.10$. Therefore, it is favorable for obtaining high resolution of the partial image range and shortened total track length of the photographing optical lens assembly.

According to the photographing optical lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic materials. When the lens elements are made of glass materials, the distribution of the refractive power of the photographing optical lens assembly may be more flexible to design. When the lens elements are made of plastic materials, manufacturing costs can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than a spherical surface so as to have more controllable variables for eliminating aberrations thereof, and to further decrease the required amount of lens elements in the photographing optical lens assembly. Therefore, the total track length of the photographing optical lens assembly can also be reduced.

According to the photographing optical lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axial region. The paraxial region refers to the region of the surface where light rays travel close to an optical axis, and the off-axial region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface can be convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface can be concave in the paraxial region thereof.

According to the photographing optical lens assembly of the present disclosure, the refractive power or the focal length of a lens element being positive or negative may refer to the refractive power or the focal length in a paraxial region of the lens element.

According to the photographing optical lens assembly of the present disclosure, the photographing optical lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the photographing optical lens assembly of the present disclosure, the image surface of the photographing optical lens assembly, based on the corresponding image sensor, can be flat or curved. In particular, the image surface can be a curved surface being concave facing towards the object side.

According to the photographing optical lens assembly of the present disclosure, an aperture stop can be configured as a middle stop. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the photographing optical lens assembly and thereby provides a wider field of view for the same.

According to the photographing optical lens assembly of the present disclosure, a non-axial point is a critical point of the lens surface where its tangent is perpendicular to an optical axis.

According to the photographing optical lens assembly of the present disclosure, the photographing optical lens assembly can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, smart TVs, surveillance systems, motion sensing input devices, driving recording systems, rearview camera systems, and wearable devices.

According to the present disclosure, an image capturing device is provided. The image capturing device includes the aforementioned photographing optical lens assembly and an image sensor, wherein the image sensor is disposed on the image side of the aforementioned photographing optical lens assembly, that is, the image sensor can be disposed on or near the image surface of the aforementioned photographing optical lens assembly. By the arrangement of the aforementioned photographing optical lens assembly, the first lens element has positive refractive power and the sixth lens element includes inflection point, so that the main converging ability of the incident light can be provided, so that the volume of the photographing optical lens assembly can be effectively reduced so as to satisfy the demand of compact size. Further, aberrations in the off-axial region can be corrected and the back focal length can be reduced so as to obtain the balance between the image quality and the volume. Preferably, the image capturing device can further include a barrel member, a holder member or a combination thereof.

According to the present disclosure, an electronic device is provided, which includes the aforementioned image capturing device. Therefore, image quality of the electronic device can be improved. Preferably, the electronic device can further include but not limited to a control unit, a display, a storage unit, a random access memory unit (RAM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-14th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment. In FIG. 1, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor 195. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, a seventh lens element 170, an IR-cut filter 180 and an image surface 190, wherein the image sensor 195 is disposed on the image surface 190 of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (110-170).

The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element 110 is made of a plastic material, and has the object-side surface 111 and the image-side surface 112 being both aspheric.

The second lens element 120 with negative refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being concave in a paraxial region thereof. The second lens element 120 is made of a plastic material, and has the object-side surface 121 and the image-side surface 122 being both aspheric.

The third lens element 130 with positive refractive power has an object-side surface 131 being convex in a paraxial region thereof and an image-side surface 132 being concave in a paraxial region thereof. The third lens element 130 is made of a plastic material, and has the object-side surface 131 and the image-side surface 132 being both aspheric.

The fourth lens element 140 with positive refractive power has an object-side surface 141 being concave in a paraxial region thereof and an image-side surface 142 being convex in a paraxial region thereof. The fourth lens element 140 is made of a plastic material, and has the object-side surface 141 and the image-side surface 142 being both aspheric. Furthermore, the image-side surface 142 of the fourth lens element 140 includes at least one inflection point.

The fifth lens element 150 with negative refractive power has an object-side surface 151 being concave in a paraxial region thereof and an image-side surface 152 being convex in a paraxial region thereof. The fifth lens element 150 is made of a plastic material, and has the object-side surface 151 and the image-side surface 152 being both aspheric.

The sixth lens element 160 with negative refractive power has an object-side surface 161 being concave in a paraxial region thereof and an image-side surface 162 being concave in a paraxial region thereof. The sixth lens element 160 is made of a plastic material, and has the object-side surface 161 and the image-side surface 162 being both aspheric. Furthermore, the image-side surface 162 of the sixth lens element 160 includes at least one inflection point.

The seventh lens element 170 with positive refractive power has an object-side surface 171 being concave in a paraxial region thereof and an image-side surface 172 being convex in a paraxial region thereof. The seventh lens element 170 is made of a plastic material, and has the object-side surface 171 and the image-side surface 172 being both aspheric.

The IR-cut filter 180 is made of a glass material and located between the seventh lens element 170 and the image surface 190, and will not affect the focal length of the photographing optical lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the photographing optical lens assembly according to the 1st embodiment, when a focal length of the photograph-

ing optical lens assembly is f , an f -number of the photographing optical lens assembly is Fno , and half of a maximal field of view of the photographing optical lens assembly is $HFOV$, these parameters have the following values: $f=6.17$ mm; $Fno=2.40$; and $HFOV=20.8$ degrees.

In the photographing optical lens assembly according to the 1st embodiment, when an Abbe number of the seventh lens element **170** is $V7$, the following condition is satisfied: $V7=23.5$.

In the photographing optical lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface **111** of the first lens element **110** is $R1$, and a central thickness of the first lens element **110** is $CT1$, the following condition is satisfied: $R1/CT1=1.51$.

In the photographing optical lens assembly according to the 1st embodiment, when an axial distance between the first lens element **110** and the second lens element **120** is $T12$, an axial distance between the second lens element **120** and the third lens element **130** is $T23$, an axial distance between the third lens element **130** and the fourth lens element **140** is $T34$, an axial distance between the fourth lens element **140** and the fifth lens element **150** is $T45$, an axial distance between the fifth lens element **150** and the sixth lens element **160** is $T56$, an axial distance between the sixth lens element **160** and the seventh lens element **170** is $T67$, and a sum of axial distances between every two of the lens elements of the photographing optical lens assembly that are adjacent to each other is ΣAT (that is, $\Sigma AT=T12+T23+T34+T45+T56+T67$), the following condition is satisfied: $T56/(\Sigma AT-T56)=0.68$.

In the photographing optical lens assembly according to the 1st embodiment, when the focal length of the photographing optical lens assembly is f , the curvature radius of the object-side surface **111** of the first lens element **110** is $R1$, and a curvature radius of the image-side surface **172** of the seventh lens element **170** is $R14$, the following conditions are satisfied: $f/R1=3.79$; and $f/R14=-0.28$.

In the photographing optical lens assembly according to the 1st embodiment, when a focal length of the sixth lens element **160** is $f6$, and a focal length of the seventh lens element **170** is $f7$, the following condition is satisfied: $f6/f7=-0.07$.

In the photographing optical lens assembly according to the 1st embodiment, when the focal length of the photographing optical lens assembly is f , and a focal length of the fourth lens element **140** is $f4$, the following condition is satisfied: $|f/f4|=0.06$.

In the photographing optical lens assembly according to the 1st embodiment, when a refractive power of the first lens element **110** is $P1$ (which is $f/f1$, a ratio value of the focal length of the photographing optical lens assembly f and the focal length of the first lens element $f1$), a refractive power of the second lens element **120** is $P2$ (which is $f/f2$, a ratio value of the focal length of the photographing optical lens assembly f and the focal length of the second lens element $f2$), a refractive power of the third lens element **130** is $P3$ (which is $f/f3$, a ratio value of the focal length of the

photographing optical lens assembly f and the focal length of the third lens element $f3$), a refractive power of the fourth lens element **140** is $P4$ (which is $f/f4$, a ratio value of the focal length of the photographing optical lens assembly f and the focal length of the fourth lens element $f4$), a refractive power of the fifth lens element **150** is $P5$ (which is $f/f5$, a ratio value of the focal length of the photographing optical lens assembly f and the focal length of the fifth lens element $f5$), a refractive power of the sixth lens element **160** is $P6$ (which is $f/f6$, a ratio value of the focal length of the photographing optical lens assembly f and the focal length of the sixth lens element $f6$), and a refractive power of the seventh lens element **170** is $P7$ (which is $f/f7$, a ratio value of the focal length of the photographing optical lens assembly f and the focal length of the seventh lens element $f7$), the following condition is satisfied: $(|P3|+|P4|+|P5|+|P7|)/(|P1|+|P2|+|P6|)=0.06$.

FIG. **23** shows a schematic view of a parameter $Yc62$ according to the 1st embodiment of FIG. **1**. In FIG. **23**, when a vertical distance between a non-axial critical point on the image-side surface **162** of the sixth lens element **160** and an optical axis is $Yc62$, and a central thickness of the sixth lens element **160** is $CT6$, the following condition is satisfied: $Yc62/CT6=2.32$.

FIG. **24** shows a schematic view of a parameter $Dr1s$ according to the 1st embodiment of FIG. **1**. In FIG. **24**, when an axial distance between the object-side surface **111** of the first lens element **110** and the aperture stop **100** is $Dr1s$, and a central thickness of the second lens element **120** is $CT2$, the following condition is satisfied: $|Dr1s|/CT2=2.99$.

In the photographing optical lens assembly according to the 1st embodiment, when an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **190** is TL , and a focal length of the photographing optical lens assembly is f , the following condition is satisfied: $TL/f=1.00$.

In the photographing optical lens assembly according to the 1st embodiment, when the focal length of the photographing optical lens assembly is f , and a maximal image height of the photographing optical lens assembly is $ImgH$ (half of a diagonal length of an effective photosensitive area of the image sensor **195**), the following condition is satisfied: $f/ImgH=2.54$.

FIG. **25** shows a schematic view of a parameter $Y11$ according to the 1st embodiment of FIG. **1**. In FIG. **25**, when a maximal optical effective radius of the object-side surface **111** of the first lens element **110** is $Y11$, and the maximal image height of the photographing optical lens assembly is $ImgH$, the following condition is satisfied: $Y11/ImgH=0.53$.

In the photographing optical lens assembly according to the 1st embodiment, when an axial distance between the image-side surface **172** of the seventh lens element **170** and the image surface **190** is BL , and the maximal image height of the photographing optical lens assembly is $ImgH$, the following condition is satisfied: $BL/ImgH=0.28$.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment								
$f = 6.17$ mm, $Fno = 2.40$, $HFOV = 20.8$ deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.658					
2	Lens 1	1.627 ASP	1.080	Plastic	1.544	56.0	3.25	
3		15.863 ASP	0.158					

TABLE 1-continued

1st Embodiment								
f = 6.17 mm, Fno = 2.40, HFOV = 20.8 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
4	Lens 2	15.023 ASP	0.220	Plastic	1.639	23.5	-5.59	
5		2.868 ASP	0.398					
6	Lens 3	9.402 ASP	0.364	Plastic	1.544	56.0	75.46	
7		12.029 ASP	0.277					
8	Lens 4	-11.303 ASP	0.497	Plastic	1.639	23.5	95.49	
9		-9.699 ASP	0.056					
10	Lens 5	-10.442 ASP	0.638	Plastic	1.639	23.5	-3623.55	
11		-10.739 ASP	0.728					
12	Lens 6	-11.380 ASP	0.250	Plastic	1.544	56.0	-5.82	
13		4.417 ASP	0.175					
14	Lens 7	-36.851 ASP	0.643	Plastic	1.639	23.5	81.37	
15		-21.707 ASP	0.300					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.167					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-1.0386E-01	8.9967E+01	-7.3778E+01	-1.6312E+00	3.4735E+01	4.8544E+01	8.0225E+01
A4 =	5.3232E-03	-2.2658E-02	-1.0686E-01	-4.6145E-02	-1.0251E-02	-2.6165E-02	-3.4636E-02
A6 =	-3.6531E-03	-9.2838E-03	1.9523E-01	2.7362E-01	6.4442E-02	5.1237E-02	7.2759E-03
A8 =	1.4474E-02	2.0467E-01	1.2494E-01	-6.5690E-02	1.2376E-01	6.0949E-02	9.9445E-03
A10 =	-1.1366E-02	-3.3364E-01	-5.5251E-01	-1.2478E-01	-2.3471E-01	-1.2296E-01	-2.7364E-03
A12 =	4.2658E-03	2.3534E-01	4.9260E-01	7.5975E-02	1.5388E-01	8.6428E-02	
A14 =		-6.5551E-02	-1.5826E-01		-3.8658E-02	-2.5005E-02	

	Surface #						
	9	10	11	12	13	14	15
k =	-4.3662E+01	-4.7885E+00	4.3254E+01	2.1460E+01	7.5009E-01	8.8441E+01	8.2867E+01
A4 =	1.3868E-02	1.5626E-02	-1.5629E-02	-1.6221E-01	-1.9198E-01	-9.4455E-02	-8.5577E-02
A6 =	1.0986E-03	4.2101E-03	2.3887E-03	-3.2820E-02	4.0272E-02	9.6827E-02	7.3228E-02
A8 =	1.2111E-03	-9.6101E-05	3.0543E-03	7.8243E-02	3.1591E-02	-4.9012E-02	-3.2148E-02
A10 =	1.4658E-04	3.8246E-04	4.5234E-04	-3.5566E-02	-2.7562E-02	1.3964E-02	8.0531E-03
A12 =				3.2525E-03	8.8240E-03	-2.3348E-03	-1.1667E-03
A14 =				1.6709E-03	-1.3737E-03	2.2331E-04	9.2268E-05
A16 =				-3.2376E-04	9.0138E-05	-9.7154E-06	-3.0738E-06

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-18 represent the surfaces sequentially arranged from the object side to the image side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment correspond to schematic parameter and aberration curves of each embodiment, and term definitions of the tables are the same as those in Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

Furthermore, in the 1st embodiment, among the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, the sixth lens element **160** and the seventh lens element **170**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the fourth lens element **140** and the Abbe number of the seventh lens element **170** are smaller than 25.

2nd Embodiment

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment. In FIG. 3, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **295**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **200**, a first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens element **250**, a sixth lens element **260**, a seventh lens element **270**, an IR-cut filter **280** and an image surface **290**, wherein the image sensor **295** is disposed on the image surface **290** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**210-270**).

The first lens element **210** with positive refractive power has an object-side surface **211** being convex in a paraxial

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region thereof and an image-side surface **212** being concave in a paraxial region thereof. The first lens element **210** is made of a plastic material, and has the object-side surface **211** and the image-side surface **212** being both aspheric.

The second lens element **220** with negative refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof. The second lens element **220** is made of a plastic material, and has the object-side surface **221** and the image-side surface **222** being both aspheric.

The third lens element **230** with negative refractive power has an object-side surface **231** being convex in a paraxial region thereof and an image-side surface **232** being concave in a paraxial region thereof. The third lens element **230** is made of a plastic material, and has the object-side surface **231** and the image-side surface **232** being both aspheric.

The fourth lens element **240** with negative refractive power has an object-side surface **241** being concave in a paraxial region thereof and an image-side surface **242** being convex in a paraxial region thereof. The fourth lens element **240** is made of a plastic material, and has the object-side surface **241** and the image-side surface **242** being both aspheric. Furthermore, the image-side surface **242** of the fourth lens element **240** includes at least one inflection point.

The fifth lens element **250** with positive refractive power has an object-side surface **251** being concave in a paraxial

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region thereof and an image-side surface **252** being convex in a paraxial region thereof. The fifth lens element **250** is made of a plastic material, and has the object-side surface **251** and the image-side surface **252** being both aspheric.

The sixth lens element **260** with negative refractive power has an object-side surface **261** being concave in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof. The sixth lens element **260** is made of a plastic material, and has the object-side surface **261** and the image-side surface **262** being both aspheric. Furthermore, the image-side surface **262** of the sixth lens element **260** includes at least one inflection point.

The seventh lens element **270** with positive refractive power has an object-side surface **271** being concave in a paraxial region thereof and an image-side surface **272** being convex in a paraxial region thereof. The seventh lens element **270** is made of a plastic material, and has the object-side surface **271** and the image-side surface **272** being both aspheric.

The IR-cut filter **280** is made of a glass material and located between the seventh lens element **270** and the image surface **290**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 5.89 mm, Fno = 2.30, HFOV = 21.8 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.639					
2	Lens 1	1.634 ASP	1.120	Plastic	1.544	56.0	3.02	
3		175.187 ASP	0.117					
4	Lens 2	15.805 ASP	0.220	Plastic	1.639	23.5	-5.65	
5		2.920 ASP	0.306					
6	Lens 3	26.089 ASP	0.383	Plastic	1.544	56.0	-68.07	
7		15.227 ASP	0.180					
8	Lens 4	-10.643 ASP	0.655	Plastic	1.639	23.5	-29.62	
9		-24.919 ASP	0.052					
10	Lens 5	-63.172 ASP	0.935	Plastic	1.583	30.2	24.29	
11		-11.637 ASP	0.645					
12	Lens 6	-14.640 ASP	0.261	Plastic	1.544	56.0	-5.80	
13		4.049 ASP	0.171					
14	Lens 7	-51.045 ASP	0.515	Plastic	1.639	23.5	66.49	
15		-23.270 ASP	0.200					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.153					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-1.1170E-01	9.0000E+01	-7.3778E+01	-4.3908E+00	-9.0000E+01	3.3937E+01	6.8985E+01
A4 =	7.1118E-03	-1.3279E-02	-1.0228E-01	-5.4633E-02	-2.6325E-02	-2.4901E-02	-3.1142E-02
A6 =	-6.2390E-03	-3.0305E-03	1.9558E-01	2.9085E-01	7.7658E-02	3.9355E-02	8.2221E-03
A8 =	1.4620E-02	1.9792E-01	1.3328E-01	-6.2387E-02	1.2416E-01	6.4216E-02	4.2860E-03
A10 =	-1.0119E-02	-3.3418E-01	-5.4973E-01	-1.1875E-01	-2.3898E-01	-1.2873E-01	-4.9774E-04
A12 =	3.4763E-03	2.3455E-01	4.8326E-01	8.6366E-02	1.5005E-01	7.8054E-02	
A14 =		-6.3038E-02	-1.5111E-01		-3.8373E-02	-2.4073E-02	

TABLE 4-continued

	Aspheric Coefficients						
	Surface #						
	9	10	11	12	13	14	15
k =	-1.6766E+01	8.2704E+01	3.5947E+01	4.9847E+01	1.4468E+00	-9.0000E+01	9.0000E+01
A4 =	1.3429E-02	5.2861E-03	-1.9970E-02	-1.5894E-01	-1.9281E-01	-9.5274E-02	-7.4727E-02
A6 =	1.2215E-03	4.4770E-03	2.0401E-03	-2.8627E-02	3.9930E-02	9.4441E-02	6.8192E-02
A8 =	9.8190E-04	1.1306E-03	1.5536E-03	7.8240E-02	3.2058E-02	-4.9274E-02	-3.2200E-02
A10 =	1.3405E-03	-1.2579E-04	-7.4957E-05	-3.5285E-02	-2.7472E-02	1.3949E-02	8.0756E-03
A12 =				3.2490E-03	8.8365E-03	-2.3331E-03	-1.1629E-03
A14 =				1.6216E-03	-1.3765E-03	2.2428E-04	9.2500E-05
A16 =				-3.4136E-04	8.6760E-05	-9.3757E-06	-3.1172E-06

In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	5.89 f/f4		0.20
Fno	2.30 (P3 + P4 + P5 + P7)/(P1 + P2 + P6)		0.15
HFOV [deg.]	21.8 Yc62/CT6		2.38
V7	23.5 Dr1s /CT2		2.90
R1/CT1	1.46 TL/f		1.04
T56/(ΣAT - T56)	0.78 f/ImgH		2.42
f/R1	3.60 Y11/ImgH		0.53
f/R14	-0.25 BL/ImgH		0.23
f6/f7	-0.09		

Furthermore, in the 2nd embodiment, among the first lens element **210**, the second lens element **220**, the third lens element **230**, the fourth lens element **240**, the fifth lens element **250**, the sixth lens element **260** and the seventh lens element **270**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the seventh lens element **270** is smaller than 25.

3rd Embodiment

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment. In FIG. 5, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **395**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **300**, a first lens element **310**, a second lens element **320**, a third lens element **330**, a fourth lens element **340**, a fifth lens element **350**, a sixth lens element **360**, a seventh lens element **370**, an IR-cut filter **380** and an image surface **390**, wherein the image sensor **395** is disposed on the image surface **390** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**310-370**).

The first lens element **310** with positive refractive power has an object-side surface **311** being convex in a paraxial

region thereof and an image-side surface **312** being convex in a paraxial region thereof. The first lens element **310** is made of a plastic material, and has the object-side surface **311** and the image-side surface **312** being both aspheric.

The second lens element **320** with negative refractive power has an object-side surface **321** being convex in a paraxial region thereof and an image-side surface **322** being concave in a paraxial region thereof. The second lens element **320** is made of a plastic material, and has the object-side surface **321** and the image-side surface **322** being both aspheric.

The third lens element **330** with negative refractive power has an object-side surface **331** being convex in a paraxial region thereof and an image-side surface **332** being concave in a paraxial region thereof. The third lens element **330** is made of a plastic material, and has the object-side surface **331** and the image-side surface **332** being both aspheric.

The fourth lens element **340** with negative refractive power has an object-side surface **341** being concave in a paraxial region thereof and an image-side surface **342** being concave in a paraxial region thereof. The fourth lens element **340** is made of a plastic material, and has the object-side surface **341** and the image-side surface **342** being both aspheric.

The fifth lens element **350** with positive refractive power has an object-side surface **351** being convex in a paraxial region thereof and an image-side surface **352** being convex in a paraxial region thereof. The fifth lens element **350** is made of a plastic material, and has the object-side surface **351** and the image-side surface **352** being both aspheric.

The sixth lens element **360** with negative refractive power has an object-side surface **361** being concave in a paraxial region thereof and an image-side surface **362** being concave in a paraxial region thereof. The sixth lens element **360** is made of a plastic material, and has the object-side surface **361** and the image-side surface **362** being both aspheric. Furthermore, the image-side surface **362** of the sixth lens element **360** includes at least one inflection point.

The seventh lens element **370** with negative refractive power has an object-side surface **371** being convex in a paraxial region thereof and an image-side surface **372** being concave in a paraxial region thereof. The seventh lens element **370** is made of a plastic material, and has the object-side surface **371** and the image-side surface **372** being both aspheric.

The IR-cut filter **380** is made of a glass material and located between the seventh lens element **370** and the image surface **390**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment f = 5.75 mm, Fno = 2.35, HFOV = 22.3 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.550				
2	Lens 1	1.675 ASP	1.031	Plastic	1.544	55.9	3.03
3		-89.251 ASP	0.112				
4	Lens 2	13.253 ASP	0.220	Plastic	1.639	23.5	-5.78
5		2.867 ASP	0.259				
6	Lens 3	19.803 ASP	0.412	Plastic	1.544	55.9	-24.46
7		7.902 ASP	0.132				
8	Lens 4	-108.983 ASP	0.582	Plastic	1.660	20.4	-34.85
9		29.211 ASP	0.050				
10	Lens 5	13.490 ASP	1.328	Plastic	1.514	56.8	14.24
11		-15.441 ASP	0.489				
12	Lens 6	-78.014 ASP	0.374	Plastic	1.514	56.8	-5.56
13		2.969 ASP	0.084				
14	Lens 7	16.547 ASP	0.606	Plastic	1.660	20.4	-165.38
15		14.159 ASP	0.200				
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
17		Plano	0.155				
18	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-9.2782E-02	-9.0000E+01	-7.3778E+01	-6.9228E+00	-2.3179E+01	2.1975E+01	-9.0000E+01
A4 =	8.7125E-03	-5.3104E-03	-9.9997E-02	-6.0818E-02	-4.4294E-02	-3.0941E-02	-2.6123E-02
A6 =	-5.4834E-03	-2.0525E-03	1.9632E-01	2.9494E-01	7.3640E-02	2.9239E-02	1.1241E-02
A8 =	1.4027E-02	1.9304E-01	1.3670E-01	-5.7259E-02	1.1698E-01	5.9661E-02	4.7749E-03
A10 =	-9.6050E-03	-3.3362E-01	-5.4887E-01	-1.3003E-01	-2.4088E-01	-1.3272E-01	-4.2512E-03
A12 =	3.6229E-03	2.3879E-01	4.7763E-01	8.1992E-02	1.5426E-01	7.6131E-02	
A14 =		-6.5074E-02	-1.4801E-01		-4.6144E-02	-2.0880E-02	
	Surface #						
	9	10	11	12	13	14	15
k =	-9.0000E+01	-7.3255E+01	6.0976E+01	-9.0000E+01	3.2526E-01	5.6045E+01	-9.0000E+01
A4 =	-6.7893E-03	-1.3151E-02	-3.7885E-02	-1.6601E-01	-1.8399E-01	-9.8927E-02	-8.3537E-02
A6 =	-6.0032E-04	-2.2157E-03	5.0093E-03	-3.2044E-02	3.9624E-02	9.2516E-02	6.9279E-02
A8 =	4.2243E-04	3.6867E-03	1.6701E-03	8.0621E-02	3.1846E-02	-4.9314E-02	-3.2218E-02
A10 =	3.7237E-03	1.2106E-03	3.0316E-05	-3.4754E-02	-2.7449E-02	1.3963E-02	8.0566E-03
A12 =				3.1866E-03	8.8473E-03	-2.3311E-03	-1.1653E-03
A14 =				1.5530E-03	-1.3762E-03	2.2457E-04	9.2499E-05
A16 =				-3.5289E-04	8.5499E-05	-9.2682E-06	-3.0202E-06

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	5.75	f/f4	0.16
Fno	2.35	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.21
HFOV [deg.]	22.3	Yc62/CT6	2.19

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-continued

3rd Embodiment			
V7	20.4	Dr1s /CT2	2.50
R1/CT1	1.62	TL/f	1.09
T56/(ΣAT - T56)	0.77	f/ImgH	2.36
f/R1	3.43	Y11/ImgH	0.51
f/R14	0.41	BL/ImgH	0.23
f6/f7	0.03		

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4th Embodiment

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device

according to the 4th embodiment. In FIG. 7, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor 495. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, a seventh lens element 470, an IR-cut filter 480 and an image surface 490, wherein the image sensor 495 is disposed on the image surface 490 of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (410-470).

The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of a plastic material, and has the object-side surface 411 and the image-side surface 412 being both aspheric.

The second lens element 420 with negative refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being concave in a paraxial region thereof. The second lens element 420 is made of a plastic material, and has the object-side surface 421 and the image-side surface 422 being both aspheric.

The third lens element 430 with positive refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof. The third lens element 430 is made of a plastic material, and has the object-side surface 431 and the image-side surface 432 being both aspheric.

The fourth lens element 440 with positive refractive power has an object-side surface 441 being concave in a

paraxial region thereof and an image-side surface 442 being convex in a paraxial region thereof. The fourth lens element 440 is made of a plastic material, and has the object-side surface 441 and the image-side surface 442 being both aspheric. Furthermore, the object-side surface 441 of the fourth lens element 440 includes at least one inflection point.

The fifth lens element 450 with negative refractive power has an object-side surface 451 being concave in a paraxial region thereof and an image-side surface 452 being convex in a paraxial region thereof. The fifth lens element 450 is made of a plastic material, and has the object-side surface 451 and the image-side surface 452 being both aspheric.

The sixth lens element 460 with negative refractive power has an object-side surface 461 being concave in a paraxial region thereof and an image-side surface 462 being concave in a paraxial region thereof. The sixth lens element 460 is made of a plastic material, and has the object-side surface 461 and the image-side surface 462 being both aspheric. Furthermore, the image-side surface 462 of the sixth lens element 460 includes at least one inflection point.

The seventh lens element 470 with positive refractive power has an object-side surface 471 being concave in a paraxial region thereof and an image-side surface 472 being convex in a paraxial region thereof. The seventh lens element 470 is made of a plastic material, and has the object-side surface 471 and the image-side surface 472 being both aspheric.

The IR-cut filter 480 is made of a glass material and located between the seventh lens element 470 and the image surface 490, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment								
f = 6.14 mm, Fno = 2.40, HFOV = 17.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.626					
2	Lens 1	1.658 ASP	1.143	Plastic	1.544	55.9	3.27	
3		18.459 ASP	0.162					
4	Lens 2	10.137 ASP	0.221	Plastic	1.639	23.5	-5.43	
5		2.562 ASP	0.409					
6	Lens 3	8.296 ASP	0.278	Plastic	1.544	55.9	147.80	
7		9.140 ASP	0.209					
8	Lens 4	-22.557 ASP	0.343	Plastic	1.515	56.5	25.53	
9		-8.346 ASP	0.075					
10	Lens 5	-9.965 ASP	1.385	Plastic	1.639	23.5	-47.89	
11		-15.580 ASP	0.527					
12	Lens 6	-11.655 ASP	0.220	Plastic	1.544	55.9	-5.53	
13		4.081 ASP	0.128					
14	Lens 7	-89.561 ASP	0.572	Plastic	1.639	23.5	38.29	
15		-19.257 ASP	0.250					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.174					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients							
Surface #							
	2	3	4	5	6	7	8
k =	-1.0876E-01	9.0000E+01	-7.3778E+01	-1.7184E+00	3.2928E+01	5.8938E+01	-9.0000E+01
A4 =	6.3881E-03	-1.9993E-02	-1.0355E-01	-4.6342E-02	-1.1821E-02	-2.4966E-02	-2.8876E-02

TABLE 8-continued

Aspheric Coefficients						
A6 =	-5.4449E-03	-5.1561E-03	1.9674E-01	2.8592E-01	6.6552E-02	4.8491E-02
A8 =	1.5028E-02	2.0517E-01	1.2800E-01	-6.4955E-02	1.2250E-01	6.5103E-02
A10 =	-1.0600E-02	-3.3532E-01	-5.5308E-01	-1.2908E-01	-2.3644E-01	-1.2279E-01
A12 =	3.6702E-03	2.3554E-01	4.8720E-01	6.8372E-02	1.5028E-01	8.4166E-02
A14 =		-6.4952E-02	-1.5671E-01		-4.0064E-02	-2.5850E-02
						-8.8732E-07
Surface #						
	9	10	11	12	13	14
k =	-3.5090E+00	5.7860E+01	5.3515E+01	-2.3048E+01	2.4639E+00	-9.0000E+01
A4 =	5.7925E-03	7.7652E-03	-1.2043E-02	-1.4465E-01	-1.7809E-01	-1.0784E-01
A6 =	-5.0037E-03	2.5649E-03	2.6457E-03	-2.6528E-02	3.9035E-02	9.5724E-02
A8 =	-7.4716E-04	-1.1179E-03	3.0671E-03	7.9350E-02	3.1721E-02	-4.9281E-02
A10 =	-1.0889E-03	-1.5413E-04	3.6669E-04	-3.5411E-02	-2.7526E-02	1.3939E-02
A12 =	-7.5758E-04	2.9055E-04	-1.9339E-05	3.2380E-03	8.8173E-03	-2.3315E-03
A14 =	-3.0188E-04	1.4403E-04	5.4237E-05	1.6363E-03	-1.3804E-03	2.2581E-04
A16 =				-3.5363E-04	8.6976E-05	-8.9058E-06
						-3.2249E-06

In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	6.14	f/f4	0.24
Fno	2.40	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.14
HFOV [deg.]	17.5	Yc62/CT6	3.00
V7	23.5	Dr1s /CT2	2.83
R1/CT1	1.45	TL/f	1.03
T56/(ΣAT - T56)	0.54	f/ImgH	3.07
f/R1	3.70	Y11/ImgH	0.64
f/R14	-0.32	BL/ImgH	0.32
f6/f7	-0.14		

Furthermore, in the 4th embodiment, among the first lens element **410**, the second lens element **420**, the third lens element **430**, the fourth lens element **440**, the fifth lens element **450**, the sixth lens element **460** and the seventh lens element **470**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the seventh lens element **470** is smaller than 25.

5th Embodiment

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment. In FIG. 9, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **595**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **500**, a first lens element **510**, a second lens

element **520**, a third lens element **530**, a fourth lens element **540**, a fifth lens element **550**, a sixth lens element **560**, a seventh lens element **570**, an IR-cut filter **580** and an image surface **590**, wherein the image sensor **595** is disposed on the image surface **590** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**510-570**).

The first lens element **510** with positive refractive power has an object-side surface **511** being convex in a paraxial region thereof and an image-side surface **512** being convex in a paraxial region thereof. The first lens element **510** is made of a plastic material, and has the object-side surface **511** and the image-side surface **512** being both aspheric.

The second lens element **520** with negative refractive power has an object-side surface **521** being concave in a paraxial region thereof and an image-side surface **522** being concave in a paraxial region thereof. The second lens element **520** is made of a plastic material, and has the object-side surface **521** and the image-side surface **522** being both aspheric.

The third lens element **530** with negative refractive power has an object-side surface **531** being convex in a paraxial region thereof and an image-side surface **532** being concave in a paraxial region thereof. The third lens element **530** is made of a plastic material, and has the object-side surface **531** and the image-side surface **532** being both aspheric.

The fourth lens element **540** with positive refractive power has an object-side surface **541** being convex in a paraxial region thereof and an image-side surface **542** being convex in a paraxial region thereof. The fourth lens element **540** is made of a plastic material, and has the object-side surface **541** and the image-side surface **542** being both aspheric. Furthermore, each of the object-side surface **541** and the image-side surface **542** of the fourth lens element **540** includes at least one inflection point.

The fifth lens element **550** with negative refractive power has an object-side surface **551** being convex in a paraxial region thereof and an image-side surface **552** being concave in a paraxial region thereof. The fifth lens element **550** is made of a plastic material, and has the object-side surface **551** and the image-side surface **552** being both aspheric.

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The sixth lens element **560** with negative refractive power has an object-side surface **561** being concave in a paraxial region thereof and an image-side surface **562** being concave in a paraxial region thereof. The sixth lens element **560** is made of a plastic material, and has the object-side surface **561** and the image-side surface **562** being both aspheric. Furthermore, the image-side surface **562** of the sixth lens element **560** includes at least one inflection point.

The seventh lens element **570** with positive refractive power has an object-side surface **571** being concave in a paraxial region thereof and an image-side surface **572** being

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convex in a paraxial region thereof. The seventh lens element **570** is made of a plastic material, and has the object-side surface **571** and the image-side surface **572** being both aspheric.

The IR-cut filter **580** is made of a glass material and located between the seventh lens element **570** and the image surface **590**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 6.14 mm, Fno = 2.38, HFOV = 21.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.682					
2	Lens 1	1.569 ASP	1.146	Plastic	1.544	56.0	2.77	
3		-28.534 ASP	0.119					
4	Lens 2	-29.021 ASP	0.220	Plastic	1.639	23.5	-4.43	
5		3.145 ASP	0.386					
6	Lens 3	19.669 ASP	0.220	Plastic	1.544	56.0	-27.69	
7		8.498 ASP	0.262					
8	Lens 4	38.821 ASP	0.251	Plastic	1.639	23.5	56.37	
9		-493.736 ASP	0.050					
10	Lens 5	41.278 ASP	0.220	Plastic	1.639	23.5	-239.94	
11		32.450 ASP	0.837					
12	Lens 6	-10.664 ASP	0.220	Plastic	1.544	56.0	-5.47	
13		4.153 ASP	0.106					
14	Lens 7	-59.542 ASP	0.699	Plastic	1.639	23.5	33.90	
15		-15.950 ASP	0.300					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.404					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-1.2365E-01	3.2180E+01	-7.3778E+01	-2.1902E+00	8.8977E+01	4.4102E+01	-9.0000E+01
A4 =	5.8271E-03	-1.9868E-02	-1.0468E-01	-4.8086E-02	-9.1784E-03	-1.7954E-02	-3.8648E-02
A6 =	-6.3763E-03	-6.3534E-03	1.9687E-01	2.8022E-01	8.1007E-02	4.8278E-02	8.2441E-05
A8 =	1.4323E-02	2.0386E-01	1.3133E-01	-6.2654E-02	1.3123E-01	6.5309E-02	5.2653E-03
A10 =	-1.0639E-02	-3.3696E-01	-5.4802E-01	-1.0740E-01	-2.3367E-01	-1.2345E-01	-3.0718E-04
A12 =	3.6856E-03	2.3524E-01	4.8783E-01	9.6195E-02	1.5277E-01	8.3907E-02	1.6370E-03
A14 =		-6.2900E-02	-1.5267E-01		-3.9541E-02	-2.5082E-02	-9.2505E-04
	Surface #						
	9	10	11	12	13	14	15
k =	-4.5000E+01	-9.0000E+01	7.9480E+01	3.3324E+01	1.3847E+00	-1.1375E+01	5.6056E+01
A4 =	1.6073E-02	1.4331E-02	-1.2330E-02	-1.6250E-01	-1.7995E-01	-1.1048E-01	-1.1876E-01
A6 =	5.5496E-03	2.8971E-03	5.7542E-03	-3.3733E-02	3.7540E-02	9.6567E-02	7.4522E-02
A8 =	2.3053E-03	5.1482E-04	4.0339E-03	7.8941E-02	3.1167E-02	-4.8955E-02	-3.2021E-02
A10 =	2.4061E-04	7.4312E-04	5.4569E-04	-3.5880E-02	-2.7507E-02	1.3969E-02	8.0596E-03
A12 =	9.1951E-06	1.7063E-05	-1.6016E-04	2.9813E-03	8.8434E-03	-2.3342E-03	-1.1666E-03
A14 =	4.7156E-04	-4.3989E-04	-2.0317E-04	1.6142E-03	-1.3755E-03	2.2344E-04	9.2160E-05
A16 =				-2.7677E-04	8.6482E-05	-9.6525E-06	-3.1195E-06

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In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment			
f [mm]	6.14	f/f4	0.11
Fno	2.38	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.11
HFOV [deg.]	21.0	Yc62/CT6	2.91
V7	23.5	Dr1s /CT2	3.10
R1/CT1	1.37	TL/f	0.92
T56/(ΣAT - T56)	0.91	f/ImgH	2.52
f/R1	3.91	Y11/ImgH	0.53
f/R14	-0.39	BL/ImgH	0.38
f6/f7	-0.16		

Furthermore, in the 5th embodiment, among the first lens element **510**, the second lens element **520**, the third lens element **530**, the fourth lens element **540**, the fifth lens element **550**, the sixth lens element **560** and the seventh lens element **570**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, both of the Abbe number of the fourth lens element **540** and the Abbe number of the seventh lens element **570** are smaller than 25.

6th Embodiment

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure. FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment. In FIG. 11, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **695**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **600**, a first lens element **610**, a second lens element **620**, a third lens element **630**, a fourth lens element **640**, a fifth lens element **650**, a sixth lens element **660**, a seventh lens element **670**, an IR-cut filter **680** and an image surface **690**, wherein the image sensor **695** is disposed on the image surface **690** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**610-670**).

The first lens element **610** with positive refractive power has an object-side surface **611** being convex in a paraxial region thereof and an image-side surface **612** being convex

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in a paraxial region thereof. The first lens element **610** is made of a plastic material, and has the object-side surface **611** and the image-side surface **612** being both aspheric.

The second lens element **620** with negative refractive power has an object-side surface **621** being concave in a paraxial region thereof and an image-side surface **622** being concave in a paraxial region thereof. The second lens element **620** is made of a plastic material, and has the object-side surface **621** and the image-side surface **622** being both aspheric.

The third lens element **630** with negative refractive power has an object-side surface **631** being convex in a paraxial region thereof and an image-side surface **632** being concave in a paraxial region thereof. The third lens element **630** is made of a plastic material, and has the object-side surface **631** and the image-side surface **632** being both aspheric.

The fourth lens element **640** with positive refractive power has an object-side surface **641** being convex in a paraxial region thereof and an image-side surface **642** being convex in a paraxial region thereof. The fourth lens element **640** is made of a plastic material, and has the object-side surface **641** and the image-side surface **642** being both aspheric. Furthermore, each of the object-side surface **641** and the image-side surface **642** of the fourth lens element **640** includes at least one inflection point.

The fifth lens element **650** with negative refractive power has an object-side surface **651** being concave in a paraxial region thereof and an image-side surface **652** being convex in a paraxial region thereof. The fifth lens element **650** is made of a plastic material, and has the object-side surface **651** and the image-side surface **652** being both aspheric.

The sixth lens element **660** with negative refractive power has an object-side surface **661** being concave in a paraxial region thereof and an image-side surface **662** being concave in a paraxial region thereof. The sixth lens element **660** is made of a plastic material, and has the object-side surface **661** and the image-side surface **662** being both aspheric. Furthermore, the image-side surface **662** of the sixth lens element **660** includes at least one inflection point.

The seventh lens element **670** with positive refractive power has an object-side surface **671** being concave in a paraxial region thereof and an image-side surface **672** being convex in a paraxial region thereof. The seventh lens element **670** is made of a plastic material, and has the object-side surface **671** and the image-side surface **672** being both aspheric.

The IR-cut filter **680** is made of a glass material and located between the seventh lens element **670** and the image surface **690**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment								
f = 6.16 mm, Fno = 2.45, HFOV = 21.0 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.654				
2	Lens 1	1.537	ASP	1.057	Plastic	1.544	56.0	2.61
3		-14.491	ASP	0.115				
4	Lens 2	-18.580	ASP	0.220	Plastic	1.639	23.3	-4.33
5		3.269	ASP	0.441				
6	Lens 3	71.366	ASP	0.220	Plastic	1.535	55.8	-16.21

TABLE 11-continued

6th Embodiment f = 6.16 mm, Fno = 2.45, HFOV = 21.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
7		7.725	ASP	0.304			
8	Lens 4	68.089	ASP	0.259	Plastic	1.639	23.3
9		-15.202	ASP	0.124			
10	Lens 5	-12.217	ASP	0.220	Plastic	1.535	55.8
11		-155.931	ASP	0.711			-24.79
12	Lens 6	-9.848	ASP	0.220	Plastic	1.544	56.0
13		4.761	ASP	0.137			-5.87
14	Lens 7	-18.619	ASP	0.697	Plastic	1.639	23.5
15		-14.553	ASP	0.300			97.83
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
17		Plano	0.317				
18	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-1.1642E-01	-2.6582E+01	-7.3778E+01	-2.2676E+00	-8.7819E+01	4.1689E+01	-8.9694E+01
A4 =	6.6285E-03	-1.9071E-02	-1.0358E-01	-4.8163E-02	-6.1969E-03	-1.8156E-02	-3.8897E-02
A6 =	-6.4338E-03	-5.1092E-03	1.9822E-01	2.8370E-01	8.3100E-02	4.9744E-02	-2.3033E-03
A8 =	1.4333E-02	2.0430E-01	1.3372E-01	-6.1567E-02	1.3465E-01	6.5826E-02	4.6351E-03
A10 =	-1.0502E-02	-3.3810E-01	-5.4657E-01	-1.0399E-01	-2.3036E-01	-1.2390E-01	-1.3335E-03
A12 =	3.8258E-03	2.3309E-01	4.8759E-01	1.1429E-01	1.5425E-01	8.2898E-02	-2.3329E-04
A14 =		-6.1101E-02	-1.5344E-01		-4.0283E-02	-2.5792E-02	-3.2920E-03
	Surface #						
	9	10	11	12	13	14	15
k =	8.3766E+01	7.8291E+00	9.0000E+01	3.2088E+01	1.3448E+00	9.0000E+01	4.7990E+01
A4 =	1.5740E-02	1.3920E-02	-1.2978E-02	-1.7245E-01	-1.7786E-01	-1.1048E-01	-1.3706E-01
A6 =	7.0736E-03	2.0079E-03	6.6686E-03	-3.6749E-02	3.5165E-02	9.6861E-02	7.7647E-02
A8 =	2.8411E-03	4.3594E-04	4.3277E-03	7.8726E-02	3.0716E-02	-4.8853E-02	-3.1883E-02
A10 =	5.8614E-04	7.3163E-04	5.6449E-04	-3.5859E-02	-2.7489E-02	1.3985E-02	8.0432E-03
A12 =	2.3853E-04	-3.3389E-05	-2.6037E-04	3.0063E-03	8.8678E-03	-2.3320E-03	-1.1717E-03
A14 =	4.7706E-04	-4.7730E-04	-3.4795E-04	1.6628E-03	-1.3699E-03	2.2385E-04	9.1219E-05
A16 =				-2.0758E-04	8.5992E-05	-9.5409E-06	-3.2845E-06

In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	6.16	f/f4	0.32
Fno	2.45	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.21
HFOV [deg.]	21.0	Yc62/CT6	2.64
V7	23.5	Dr1s /CT2	2.97
R1/CT1	1.45	TL/f	0.90
T56/(ΣAT - T56)	0.63	f/ImgH	2.53
f/R1	4.01	Y11/ImgH	0.52
f/R14	-0.42	BL/ImgH	0.34
f6/f7	-0.06		

Furthermore, in the 6th embodiment, among the first lens element **610**, the second lens element **620**, the third lens element **630**, the fourth lens element **640**, the fifth lens element **650**, the sixth lens element **660** and the seventh lens element **670**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, both of the Abbe number of the fourth lens element **640** and the Abbe number of the seventh lens element **670** are smaller than 25.

7th Embodiment

FIG. **13** is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. **14** shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment. In FIG. **13**, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **795**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **700**, a first lens element **710**, a second lens

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element **720**, a third lens element **730**, a fourth lens element **740**, a fifth lens element **750**, a sixth lens element **760**, a seventh lens element **770**, an IR-cut filter **780** and an image surface **790**, wherein the image sensor **795** is disposed on the image surface **790** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**710-770**).

The first lens element **710** with positive refractive power has an object-side surface **711** being convex in a paraxial region thereof and an image-side surface **712** being convex in a paraxial region thereof. The first lens element **710** is made of a plastic material, and has the object-side surface **711** and the image-side surface **712** being both aspheric.

The second lens element **720** with negative refractive power has an object-side surface **721** being convex in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof. The second lens element **720** is made of a plastic material, and has the object-side surface **721** and the image-side surface **722** being both aspheric.

The third lens element **730** with negative refractive power has an object-side surface **731** being convex in a paraxial region thereof and an image-side surface **732** being concave in a paraxial region thereof. The third lens element **730** is made of a plastic material, and has the object-side surface **731** and the image-side surface **732** being both aspheric.

The fourth lens element **740** with negative refractive power has an object-side surface **741** being convex in a paraxial region thereof and an image-side surface **742** being concave in a paraxial region thereof. The fourth lens element **740** is made of a plastic material, and has the object-side

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surface **741** and the image-side surface **742** being both aspheric. Furthermore, the object-side surface **741** of the fourth lens element **740** includes at least one inflection point.

The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex in a paraxial region thereof and an image-side surface **752** being convex in a paraxial region thereof. The fifth lens element **750** is made of a plastic material, and has the object-side surface **751** and the image-side surface **752** being both aspheric.

The sixth lens element **760** with negative refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof. The sixth lens element **760** is made of a plastic material, and has the object-side surface **761** and the image-side surface **762** being both aspheric. Furthermore, each of the object-side surface **761** and the image-side surface **762** of the sixth lens element **760** includes at least one inflection point.

The seventh lens element **770** with positive refractive power has an object-side surface **771** being convex in a paraxial region thereof and an image-side surface **772** being concave in a paraxial region thereof. The seventh lens element **770** is made of a plastic material, and has the object-side surface **771** and the image-side surface **772** being both aspheric.

The IR-cut filter **780** is made of a glass material and located between the seventh lens element **770** and the image surface **790**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment								
f = 5.59 mm, Fno = 2.25, HFOV = 21.7 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.580				
2	Lens 1	1.670	ASP	1.037	Plastic	1.544	55.9	3.05
3		-186.098	ASP	0.097				
4	Lens 2	10.810	ASP	0.220	Plastic	1.639	23.5	-5.32
5		2.563	ASP	0.270				
6	Lens 3	10.819	ASP	0.477	Plastic	1.544	55.9	-25.89
7		6.025	ASP	0.096				
8	Lens 4	19.116	ASP	0.702	Plastic	1.660	20.4	-34.04
9		10.177	ASP	0.098				
10	Lens 5	7.956	ASP	1.153	Plastic	1.514	56.8	12.73
11		-34.844	ASP	0.519				
12	Lens 6	53.960	ASP	0.345	Plastic	1.514	56.8	-5.79
13		2.813	ASP	0.100				
14	Lens 7	13.418	ASP	0.442	Plastic	1.660	20.4	56.04
15		20.782	ASP	0.250				
16	IR-cut filter	Plano		0.210	Glass	1.517	64.2	—
17		Plano		0.111				
18	Image	Plano		—				

Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients							
Surface #							
	2	3	4	5	6	7	8
k =	-9.3949E-02	-9.0000E+01	-7.5260E+01	-6.9770E+00	-4.8689E+01	1.4970E+01	-8.9954E+01
A4 =	8.7527E-03	-6.6993E-04	-1.0114E-01	-6.1932E-02	-4.7422E-02	-4.0348E-02	-2.7860E-02

TABLE 14-continued

Aspheric Coefficients						
A6 =	-5.3645E-03	-1.0546E-03	1.9562E-01	2.9440E-01	7.4904E-02	2.7915E-02
A8 =	1.3677E-02	1.9013E-01	1.3836E-01	-5.5578E-02	1.1669E-01	6.0246E-02
A10 =	-9.3524E-03	-3.3489E-01	-5.4770E-01	-1.3374E-01	-2.4026E-01	-1.3397E-01
A12 =	3.4651E-03	2.4123E-01	4.7389E-01	8.7398E-02	1.5472E-01	7.4616E-02
A14 =		-6.6194E-02	-1.4589E-01		-4.4155E-02	-1.9277E-02

Surface #						
	9	10	11	12	13	14
k =	-9.0000E+01	-7.7972E+01	8.9163E+01	9.0000E+01	1.4674E-01	4.0738E+01
A4 =	-8.5408E-03	-1.6233E-02	-4.2063E-02	-1.7301E-01	-1.8507E-01	-9.4764E-02
A6 =	-1.3012E-03	-1.8593E-03	4.7163E-03	-3.3749E-02	3.7597E-02	9.1506E-02
A8 =	3.1147E-04	3.7597E-03	8.7914E-04	7.9972E-02	3.1839E-02	-4.9483E-02
A10 =	4.5623E-03	4.3346E-04	-1.1836E-04	-3.4759E-02	-2.7426E-02	1.3944E-02
A12 =	6.3928E-05			3.1569E-03	8.8544E-03	-2.3335E-03
A14 =				1.5548E-03	-1.3751E-03	2.2402E-04
A16 =				-3.3018E-04	8.5389E-05	-9.4722E-06

In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment			
f [mm]	5.59	f/f4	0.16
Fno	2.25	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.24
HFOV [deg.]	21.7	Yc62/CT6	2.43
V7	20.4	Dr1s /CT2	2.64
R1/CT1	1.61	TL/f	1.10
T56/(ΣAT - T56)	0.79	f/ImgH	2.43
f/R1	3.35	Y11/ImgH	0.54
f/R14	0.27	BL/ImgH	0.25
f6/f7	-0.10		

Furthermore, in the 7th embodiment, among the first lens element **710**, the second lens element **720**, the third lens element **730**, the fourth lens element **740**, the fifth lens element **750**, the sixth lens element **760** and the seventh lens element **770**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the seventh lens element **770** is smaller than 25.

8th Embodiment

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment. In FIG. 15, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **895**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **800**, a first lens element **810**, a second lens

element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, a sixth lens element **860**, a seventh lens element **870**, an IR-cut filter **880** and an image surface **890**, wherein the image sensor **895** is disposed on the image surface **890** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**810-870**).

The first lens element **810** with positive refractive power has an object-side surface **811** being convex in a paraxial region thereof and an image-side surface **812** being convex in a paraxial region thereof. The first lens element **810** is made of a plastic material, and has the object-side surface **811** and the image-side surface **812** being both aspheric.

The second lens element **820** with negative refractive power has an object-side surface **821** being convex in a paraxial region thereof and an image-side surface **822** being concave in a paraxial region thereof. The second lens element **820** is made of a plastic material, and has the object-side surface **821** and the image-side surface **822** being both aspheric.

The third lens element **830** with negative refractive power has an object-side surface **831** being convex in a paraxial region thereof and an image-side surface **832** being concave in a paraxial region thereof. The third lens element **830** is made of a plastic material, and has the object-side surface **831** and the image-side surface **832** being both aspheric.

The fourth lens element **840** with negative refractive power has an object-side surface **841** being convex in a paraxial region thereof and an image-side surface **842** being concave in a paraxial region thereof. The fourth lens element **840** is made of a plastic material, and has the object-side surface **841** and the image-side surface **842** being both aspheric. Furthermore, each of the object-side surface **841** and the image-side surface **842** of the fourth lens element **840** includes at least one inflection point.

The fifth lens element **850** with positive refractive power has an object-side surface **851** being convex in a paraxial region thereof and an image-side surface **852** being convex in a paraxial region thereof. The fifth lens element **850** is made of a plastic material, and has the object-side surface **851** and the image-side surface **852** being both aspheric.

The sixth lens element **860** with negative refractive power has an object-side surface **861** being convex in a paraxial

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region thereof and an image-side surface **862** being concave in a paraxial region thereof. The sixth lens element **860** is made of a plastic material, and has the object-side surface **861** and the image-side surface **862** being both aspheric. Furthermore, each of the object-side surface **861** and the image-side surface **862** of the sixth lens element **860** includes at least one inflection point.

The seventh lens element **870** with positive refractive power has an object-side surface **871** being convex in a paraxial region thereof and an image-side surface **872** being concave in a paraxial region thereof. The seventh lens

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element **870** is made of a plastic material, and has the object-side surface **871** and the image-side surface **872** being both aspheric.

The IR-cut filter **880** is made of a glass material and located between the seventh lens element **870** and the image surface **890**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment								
f = 5.83 mm, Fno = 2.25, HFOV = 19.2 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.604					
2	Lens 1	1.726 ASP	1.153	Plastic	1.544	55.9	3.09	
3		-46.733 ASP	0.097					
4	Lens 2	12.194 ASP	0.220	Plastic	1.639	23.5	-5.13	
5		2.566 ASP	0.286					
6	Lens 3	11.048 ASP	0.463	Plastic	1.544	55.9	-24.40	
7		5.942 ASP	0.093					
8	Lens 4	19.815 ASP	0.876	Plastic	1.660	20.4	-38.94	
9		10.992 ASP	0.351					
10	Lens 5	12.127 ASP	1.107	Plastic	1.514	56.8	10.86	
11		-10.009 ASP	0.367					
12	Lens 6	191.264 ASP	0.412	Plastic	1.514	56.8	-5.53	
13		2.796 ASP	0.079					
14	Lens 7	12.774 ASP	0.353	Plastic	1.660	20.4	69.65	
15		17.495 ASP	0.200					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.152					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-1.0506E-01	1.7034E+01	-7.6409E+01	-6.6863E+00	-7.5034E+01	1.3806E+01	-9.0000E+01
A4 =	7.9432E-03	-4.1498E-04	-1.0138E-01	-6.1781E-02	-5.0429E-02	-4.0778E-02	-2.8621E-02
A6 =	-5.3395E-03	-1.2838E-03	1.9484E-01	2.9249E-01	7.5539E-02	2.5246E-02	6.4687E-03
A8 =	1.3621E-02	1.8923E-01	1.3639E-01	-5.6913E-02	1.1817E-01	5.7417E-02	1.9540E-03
A10 =	-9.4855E-03	-3.3674E-01	-5.5030E-01	-1.3390E-01	-2.4269E-01	-1.3612E-01	-4.3821E-03
A12 =	3.1289E-03	2.4046E-01	4.7278E-01	8.2791E-02	1.5088E-01	7.4724E-02	8.4833E-04
A14 =		-6.3433E-02	-1.4086E-01		-4.3446E-02	-1.7843E-02	
	Surface #						
	9	10	11	12	13	14	15
k =	-7.6413E+01	-6.9105E+01	3.4336E+01	-9.0000E+01	2.2700E-01	3.6744E+01	4.0364E+01
A4 =	-1.2964E-02	-2.0597E-02	-5.5726E-02	-1.8115E-01	-1.7824E-01	-8.8514E-02	-6.8094E-02
A6 =	-2.9735E-03	-4.6472E-03	4.9694E-03	-3.0919E-02	3.6752E-02	9.2746E-02	6.4457E-02
A8 =	-6.4803E-04	2.6283E-03	1.3335E-03	7.9078E-02	3.2005E-02	-4.9716E-02	-3.1785E-02
A10 =	3.9129E-03	-9.1565E-05	-1.4832E-04	-3.4831E-02	-2.7437E-02	1.4026E-02	8.0933E-03
A12 =	-2.6466E-04			3.1985E-03	8.8495E-03	-2.3206E-03	-1.1670E-03
A14 =				1.5668E-03	-1.3750E-03	2.2181E-04	9.1185E-05
A16 =				-3.3108E-04	8.5708E-05	-1.1194E-05	-3.3679E-06

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In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	5.83	f/f4	0.15
Fno	2.25	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.25
HFOV [deg.]	19.2	Yc62/CT6	2.16
V7	20.4	Dr1s /CT2	2.75
R1/CT1	1.50	TL/f	1.10
T56/(ΣAT - T56)	0.41	f/ImgH	2.77
f/R1	3.38	Y11/ImgH	0.62
f/R14	0.33	BL/ImgH	0.27
f6/f7	-0.08		

Furthermore, in the 8th embodiment, among the first lens element **810**, the second lens element **820**, the third lens element **830**, the fourth lens element **840**, the fifth lens element **850**, the sixth lens element **860** and the seventh lens element **870**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the seventh lens element **870** is smaller than 25.

9th Embodiment

FIG. 17 is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure. FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment. In FIG. 17, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **995**. The photographing optical lens assembly includes, in order from an object side to an image side, an aperture stop **900**, a first lens element **910**, a second lens element **920**, a third lens element **930**, a fourth lens element **940**, a fifth lens element **950**, a sixth lens element **960**, a seventh lens element **970**, an IR-cut filter **980** and an image surface **990**, wherein the image sensor **995** is disposed on the image surface **990** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**910-970**).

The first lens element **910** with positive refractive power has an object-side surface **911** being convex in a paraxial region thereof and an image-side surface **912** being concave

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in a paraxial region thereof. The first lens element **910** is made of a plastic material, and has the object-side surface **911** and the image-side surface **912** being both aspheric.

The second lens element **920** with negative refractive power has an object-side surface **921** being convex in a paraxial region thereof and an image-side surface **922** being concave in a paraxial region thereof. The second lens element **920** is made of a plastic material, and has the object-side surface **921** and the image-side surface **922** being both aspheric.

The third lens element **930** with negative refractive power has an object-side surface **931** being convex in a paraxial region thereof and an image-side surface **932** being concave in a paraxial region thereof. The third lens element **930** is made of a plastic material, and has the object-side surface **931** and the image-side surface **932** being both aspheric.

The fourth lens element **940** with positive refractive power has an object-side surface **941** being convex in a paraxial region thereof and an image-side surface **942** being convex in a paraxial region thereof. The fourth lens element **940** is made of a plastic material, and has the object-side surface **941** and the image-side surface **942** being both aspheric. Furthermore, each of the object-side surface **941** and the image-side surface **942** of the fourth lens element **940** includes at least one inflection point.

The fifth lens element **950** with positive refractive power has an object-side surface **951** being convex in a paraxial region thereof and an image-side surface **952** being convex in a paraxial region thereof. The fifth lens element **950** is made of a plastic material, and has the object-side surface **951** and the image-side surface **952** being both aspheric.

The sixth lens element **960** with negative refractive power has an object-side surface **961** being convex in a paraxial region thereof and an image-side surface **962** being concave in a paraxial region thereof. The sixth lens element **960** is made of a plastic material, and has the object-side surface **961** and the image-side surface **962** being both aspheric. Furthermore, each of the object-side surface **961** and the image-side surface **962** of the sixth lens element **960** includes at least one inflection point.

The seventh lens element **970** with positive refractive power has an object-side surface **971** being convex in a paraxial region thereof and an image-side surface **972** being concave in a paraxial region thereof. The seventh lens element **970** is made of a plastic material, and has the object-side surface **971** and the image-side surface **972** being both aspheric.

The IR-cut filter **980** is made of a glass material and located between the seventh lens element **970** and the image surface **990**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

TABLE 17

9th Embodiment								
f = 5.66 mm, Fno = 2.12, HFOV = 20.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.653					
2	Lens 1	1.750	ASP	1.170	Plastic	1.544	55.9	3.23
3		297.503	ASP	0.088				
4	Lens 2	7.579	ASP	0.221	Plastic	1.650	21.4	-5.14
5		2.290	ASP	0.306				

TABLE 17-continued

9th Embodiment f = 5.66 mm, Fno = 2.12, HFOV = 20.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
6	Lens 3	8.044	ASP	0.446	Plastic	1.544	55.9	-26.23
7		5.044	ASP	0.181				
8	Lens 4	49.728	ASP	0.761	Plastic	1.660	20.4	43.25
9		-66.579	ASP	0.756				
10	Lens 5	22.241	ASP	0.679	Plastic	1.514	56.8	13.79
11		-10.288	ASP	0.231				
12	Lens 6	5.895	ASP	0.370	Plastic	1.514	56.8	-5.69
13		1.912	ASP	0.089				
14	Lens 7	2.854	ASP	0.408	Plastic	1.530	55.8	98.16
15		2.870	ASP	0.300				
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2		—
17		Plano	0.093					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 18

Aspheric Coefficients							
	Surface #						
	2	3	4	5	6	7	8
k =	-1.8456E-01	9.0000E+01	-5.5627E+01	-6.4891E+00	-8.1281E+01	1.3419E+01	9.0000E+01
A4 =	9.5549E-03	1.6127E-03	-1.0076E-01	-6.4012E-02	-5.0175E-02	-4.4111E-02	-2.6811E-02
A6 =	-4.0601E-03	4.8711E-03	1.9511E-01	2.9285E-01	7.8000E-02	2.7230E-02	4.5802E-03
A8 =	1.3607E-02	1.8492E-01	1.4061E-01	-5.6552E-02	1.1636E-01	5.7306E-02	1.3834E-03
A10 =	-9.5625E-03	-3.4165E-01	-5.5142E-01	-1.4413E-01	-2.5041E-01	-1.4146E-01	-2.0113E-03
A12 =	3.0867E-03	2.4358E-01	4.6071E-01	8.7064E-02	1.4997E-01	7.2540E-02	2.7702E-03
A14 =		-6.3579E-02	-1.3324E-01		-3.8371E-02	-1.2924E-02	
	Surface #						
	9	10	11	12	13	14	15
k =	-9.0000E+01	8.6009E+01	3.4319E+01	2.2486E-01	-3.0192E-01	-2.4604E+01	-3.3102E+01
A4 =	-2.0820E-02	-1.2347E-02	-6.1008E-02	-1.8099E-01	-1.9679E-01	-8.8605E-02	-6.3672E-02
A6 =	-3.9622E-03	-9.6049E-03	1.2736E-02	-3.0633E-02	3.1550E-02	9.2779E-02	6.5933E-02
A8 =	-4.8968E-04	2.6239E-03	1.3301E-03	7.9129E-02	3.2406E-02	-4.8639E-02	-3.1504E-02
A10 =	3.1979E-03	8.3360E-05	-5.3129E-04	-3.4713E-02	-2.7265E-02	1.4180E-02	8.0957E-03
A12 =	7.1690E-04			3.1897E-03	8.8804E-03	-2.3510E-03	-1.1581E-03
A14 =				1.5531E-03	-1.3758E-03	2.1807E-04	9.2713E-05
A16 =				-3.2969E-04	8.2922E-05	-1.0046E-05	-3.9479E-06

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In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

9th Embodiment			
f [mm]	5.66	f/f4	0.13
Fno	2.12	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.21
HFOV [deg.]	20.0	Yc62/CT6	3.08
V7	55.8	Dr1s /CT2	2.95
R1/CT1	1.50	TL/f	1.12
T56/(ΣAT - T56)	0.16	f/ImgH	2.65
f/R1	3.23	Y11/ImgH	0.63

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-continued

9th Embodiment		
f/R14	1.97	BL/ImgH
f6/f7	-0.06	

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Furthermore, in the 9th embodiment, among the first lens element **910**, the second lens element **920**, the third lens element **930**, the fourth lens element **940**, the fifth lens element **950**, the sixth lens element **960** and the seventh lens element **970**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the fourth lens element **940** is smaller than 25.

10th Embodiment

FIG. **19** is a schematic view of an image capturing device according to the 10th embodiment of the present disclosure. FIG. **20** shows spherical aberration curves, astigmatic field

curves and a distortion curve of the image capturing device according to the 10th embodiment. In FIG. 19, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor 1095. The photographing optical lens assembly includes, in order from an object side to an image side, a first lens element 1010, an aperture stop 1000, a second lens element 1020, a third lens element 1030, a fourth lens element 1040, a fifth lens element 1050, a sixth lens element 1060, a seventh lens element 1070, an IR-cut filter 1080 and an image surface 1090, wherein the image sensor 1095 is disposed on the image surface 1090 of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (1010-1070).

The first lens element 1010 with positive refractive power has an object-side surface 1011 being convex in a paraxial region thereof and an image-side surface 1012 being concave in a paraxial region thereof. The first lens element 1010 is made of a plastic material, and has the object-side surface 1011 and the image-side surface 1012 being both aspheric.

The second lens element 1020 with negative refractive power has an object-side surface 1021 being convex in a paraxial region thereof and an image-side surface 1022 being concave in a paraxial region thereof. The second lens element 1020 is made of a plastic material, and has the object-side surface 1021 and the image-side surface 1022 being both aspheric.

The third lens element 1030 with positive refractive power has an object-side surface 1031 being convex in a paraxial region thereof and an image-side surface 1032 being concave in a paraxial region thereof. The third lens element 1030 is made of a plastic material, and has the object-side surface 1031 and the image-side surface 1032 being both aspheric.

The fourth lens element 1040 with negative refractive power has an object-side surface 1041 being concave in a

paraxial region thereof and an image-side surface 1042 being convex in a paraxial region thereof. The fourth lens element 1040 is made of a plastic material, and has the object-side surface 1041 and the image-side surface 1042 being both aspheric. Furthermore, each of the object-side surface 1041 and the image-side surface 1042 of the fourth lens element 1040 includes at least one inflection point.

The fifth lens element 1050 with negative refractive power has an object-side surface 1051 being concave in a paraxial region thereof and an image-side surface 1052 being convex in a paraxial region thereof. The fifth lens element 1050 is made of a plastic material, and has the object-side surface 1051 and the image-side surface 1052 being both aspheric.

The sixth lens element 1060 with negative refractive power has an object-side surface 1061 being convex in a paraxial region thereof and an image-side surface 1062 being concave in a paraxial region thereof. The sixth lens element 1060 is made of a plastic material, and has the object-side surface 1061 and the image-side surface 1062 being both aspheric. Furthermore, each of the object-side surface 1061 and the image-side surface 1062 of the sixth lens element 1060 includes at least one inflection point.

The seventh lens element 1070 with positive refractive power has an object-side surface 1071 being convex in a paraxial region thereof and an image-side surface 1072 being convex in a paraxial region thereof. The seventh lens element 1070 is made of a plastic material, and has the object-side surface 1071 and the image-side surface 1072 being both aspheric.

The IR-cut filter 1080 is made of a glass material and located between the seventh lens element 1070 and the image surface 1090, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 10th embodiment are shown in Table 19 and the aspheric surface data are shown in Table 20 below.

TABLE 19

10th Embodiment							
f = 6.21 mm, Fno = 2.50, HFOV = 20.8 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.755 ASP	1.142	Plastic	1.535	56.3	3.66
2		13.085 ASP	0.088				
3	Ape. Stop	Plano	0.050				
4	Lens 2	10.100 ASP	0.220	Plastic	1.660	20.4	-7.47
5		3.283 ASP	0.387				
6	Lens 3	9.417 ASP	0.478	Plastic	1.544	56.0	89.51
7		11.466 ASP	0.384				
8	Lens 4	-12.319 ASP	0.674	Plastic	1.639	23.5	-64.67
9		-17.930 ASP	0.050				
10	Lens 5	-520.650 ASP	0.740	Plastic	1.639	23.5	-873.78
11		-7783.993 ASP	0.779				
12	Lens 6	8.299 ASP	0.220	Plastic	1.544	56.0	-5.80
13		2.266 ASP	0.166				
14	Lens 7	24.244 ASP	0.499	Plastic	1.639	23.5	18.55
15		-22.977 ASP	0.250				
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
17		Plano	0.104				
18	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 20

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	-1.0746E-01	8.1928E+01	8.9950E+01	-5.1425E+00	-7.5952E+01	-8.9999E+01
A4 =	8.0837E-03	-7.3250E-03	-9.8844E-02	-5.6663E-02	-1.9880E-02	-2.9322E-02
A6 =	-7.4965E-03	5.7340E-03	2.0339E-01	2.7957E-01	7.1989E-02	4.2572E-02
A8 =	1.5014E-02	2.0422E-01	1.2747E-01	-7.0137E-02	1.2662E-01	7.2625E-02
A10 =	-8.8719E-03	-3.3161E-01	-5.7272E-01	-1.5037E-01	-2.3354E-01	-1.1685E-01
A12 =	2.8427E-03	2.0592E-01	4.8812E-01	7.8052E-02	1.5366E-01	8.4308E-02
A14 =	-1.4445E-04	-5.0563E-02	-1.5878E-01	2.0227E-07	-3.9685E-02	-2.4006E-02
Surface #						
	9	10	11	12	13	14
k =	3.7124E+00	9.0000E+01	-5.3178E+01	-1.1696E+01	-1.8396E+00	3.4255E+01
A4 =	1.3692E-02	1.4749E-02	-2.1548E-02	-1.8381E-01	-2.1116E-01	-9.7854E-02
A6 =	2.0696E-03	3.0456E-03	1.0547E-03	-3.8719E-02	3.6343E-02	9.5944E-02
A8 =	6.3897E-04	-2.0002E-04	1.2468E-03	7.6968E-02	3.1849E-02	-4.9143E-02
A10 =	5.6208E-04	-4.8659E-04	-1.6777E-04	-3.5713E-02	-2.7417E-02	1.3954E-02
A12 =	5.3827E-06	5.1749E-05	3.9811E-05	3.2678E-03	8.8469E-03	-2.3340E-03
A14 =				1.7015E-03	-1.3738E-03	2.2377E-04
A16 =				-3.0176E-04	8.7990E-05	-9.5428E-06

In the 10th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 19 and Table 20 as the following values and satisfy the following conditions:

10th Embodiment			
f [mm]	6.21	f/f4	0.10
Fno	2.50	(P3 + P4 + P5 + P7)/(P1 + P2 + P6)	0.14
HFOV [deg.]	20.8	Yc62/CT6	3.59
V7	23.5	Dr1s /CT2	5.59
R1/CT1	1.54	TL/f	1.04
T56/(ΣAT - T56)	0.69	f/ImgH	2.55
f/R1	3.54	Y11/ImgH	0.56
f/R14	-0.27	BL/ImgH	0.23
f6/f7	-0.31		

Furthermore, in the 10th embodiment, among the first lens element **1010**, the second lens element **1020**, the third lens element **1030**, the fourth lens element **1040**, the fifth lens element **1050**, the sixth lens element **1060** and the seventh lens element **1070**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the seventh lens element **1070** is smaller than 25.

11th Embodiment

FIG. 21 is a schematic view of an image capturing device according to the 11th embodiment of the present disclosure. FIG. 22 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 11th embodiment. In FIG. 21, the image capturing device includes a photographing optical lens assembly (its reference numeral is omitted) and an image sensor **1195**. The photographing optical lens assembly

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includes, in order from an object side to an image side, a first lens element **1110**, an aperture stop **1100**, a second lens element **1120**, a third lens element **1130**, a fourth lens element **1140**, a fifth lens element **1150**, a sixth lens element **1160**, a seventh lens element **1170**, an IR-cut filter **1180** and an image surface **1190**, wherein the image sensor **1195** is disposed on the image surface **1190** of the photographing optical lens assembly. The photographing optical lens assembly has a total of seven lens elements (**1110-1170**).

The first lens element **1110** with positive refractive power has an object-side surface **1111** being convex in a paraxial region thereof and an image-side surface **1112** being concave in a paraxial region thereof. The first lens element **1110** is made of a plastic material, and has the object-side surface **1111** and the image-side surface **1112** being both aspheric.

The second lens element **1120** with negative refractive power has an object-side surface **1121** being convex in a paraxial region thereof and an image-side surface **1122** being concave in a paraxial region thereof. The second lens element **1120** is made of a plastic material, and has the object-side surface **1121** and the image-side surface **1122** being both aspheric.

The third lens element **1130** with positive refractive power has an object-side surface **1131** being convex in a paraxial region thereof and an image-side surface **1132** being concave in a paraxial region thereof. The third lens element **1130** is made of a plastic material, and has the object-side surface **1131** and the image-side surface **1132** being both aspheric.

The fourth lens element **1140** with negative refractive power has an object-side surface **1141** being concave in a paraxial region thereof and an image-side surface **1142** being convex in a paraxial region thereof. The fourth lens element **1140** is made of a plastic material, and has the object-side surface **1141** and the image-side surface **1142** being both aspheric. Furthermore, the image-side surface **1142** of the fourth lens element **1140** includes at least one inflection point.

The fifth lens element **1150** with positive refractive power has an object-side surface **1151** being concave in a paraxial region thereof and an image-side surface **1152** being convex in a paraxial region thereof. The fifth lens element **1150** is

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made of a plastic material, and has the object-side surface **1151** and the image-side surface **1152** being both aspheric.

The sixth lens element **1160** with negative refractive power has an object-side surface **1161** being convex in a paraxial region thereof and an image-side surface **1162** being concave in a paraxial region thereof. The sixth lens element **1160** is made of a plastic material, and has the object-side surface **1161** and the image-side surface **1162** being both aspheric. Furthermore, each of the object-side surface **1161** and the image-side surface **1162** of the sixth lens element **1160** includes at least one inflection point.

The seventh lens element **1170** with negative refractive power has an object-side surface **1171** being concave in a

paraxial region thereof and an image-side surface **1172** being concave in a paraxial region thereof. The seventh lens element **1170** is made of a plastic material, and has the object-side surface **1171** and the image-side surface **1172** being both aspheric.

The IR-cut filter **1180** is made of a glass material and located between the seventh lens element **1170** and the image surface **1190**, and will not affect the focal length of the photographing optical lens assembly.

The detailed optical data of the 11th embodiment are shown in Table 21 and the aspheric surface data are shown in Table 22 below.

TABLE 21

11th Embodiment								
f = 6.21 mm, Fno = 2.40, HFOV = 21.3 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	1.734 ASP	1.260	Plastic	1.544	56.0	3.47	
2		15.937 ASP	0.077					
3	Ape. Stop	Plano	0.051					
4	Lens 2	14.813 ASP	0.220	Plastic	1.639	23.5	-5.78	
5		2.937 ASP	0.342					
6	Lens 3	8.607 ASP	0.381	Plastic	1.544	56.0	34.06	
7		15.819 ASP	0.187					
8	Lens 4	-10.561 ASP	0.606	Plastic	1.639	23.5	-26.49	
9		-28.748 ASP	0.100					
10	Lens 5	-33.186 ASP	0.960	Plastic	1.639	23.3	24.08	
11		-10.628 ASP	0.492					
12	Lens 6	13.528 ASP	0.245	Plastic	1.544	56.0	-12.96	
13		4.605 ASP	0.360					
14	Lens 7	-130.276 ASP	0.632	Plastic	1.639	23.3	-12.73	
15		8.689 ASP	0.300					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.145					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 22

Aspheric Coefficients							
Surface #							
	1	2	4	5	6	7	8
k =	-1.1821E-01	9.0000E+01	6.3210E+01	-3.6587E+00	-1.8222E+01	-7.7800E+01	7.0582E+01
A4 =	8.3856E-03	-1.0951E-02	-9.9908E-02	-5.3171E-02	-2.3103E-02	-3.1085E-02	-3.3386E-02
A6 =	-8.4669E-03	1.0486E-03	2.0575E-01	2.8582E-01	7.2677E-02	3.2691E-02	9.7772E-03
A8 =	1.5573E-02	2.0602E-01	1.3274E-01	-6.9901E-02	1.2402E-01	6.7570E-02	5.9026E-03
A10 =	-8.9916E-03	-3.2623E-01	-5.7383E-01	-1.4633E-01	-2.3344E-01	-1.1786E-01	-5.2640E-04
A12 =	2.5213E-03	2.0520E-01	4.8349E-01	7.9727E-02	1.5923E-01	8.2836E-02	1.1248E-05
A14 =		-5.1813E-02	-1.4747E-01		-4.7421E-02	-2.8707E-02	
Surface #							
	9	10	11	12	13	14	15
k =	9.0000E+01	8.9360E+01	3.6709E+01	5.1030E+01	6.4931E-01	9.0000E+01	-1.0682E+01
A4 =	1.3056E-02	1.5035E-02	-1.9537E-02	-1.7193E-01	-1.9588E-01	-1.0398E-01	-9.1170E-02
A6 =	3.6333E-03	1.9245E-03	3.3055E-03	-3.2644E-02	3.6430E-02	9.5161E-02	7.2072E-02
A8 =	1.0059E-03	-6.0489E-05	1.4950E-03	7.6798E-02	3.1565E-02	-4.9070E-02	-3.1910E-02
A10 =	6.3894E-04	-2.0245E-04	-4.4958E-04	-3.5670E-02	-2.7568E-02	1.3989E-02	8.0381E-03
A12 =	3.7850E-04	1.0643E-04		3.2889E-03	8.8338E-03	-2.3282E-03	-1.1708E-03
A14 =				1.6756E-03	-1.3640E-03	2.2424E-04	9.2140E-05
A16 =				-3.3006E-04	9.4347E-05	-9.6497E-06	-2.9895E-06

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In the 11th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 11th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 21 and Table 22 as the following values and satisfy the following conditions:

11th Embodiment			
f [mm]	6.21	$ f/f4 $	0.23
Fno	2.40	$(P3 + P4 + P5 + P7)/(P1 + P2 + P6)$	0.35
HFOV [deg.]	21.3	$Yc62/CT6$	2.29
V7	23.3	$ Dr1s /CT2$	6.08
R1/CT1	1.38	TL/f	1.06
$T56/(\Sigma AT - T56)$	0.44	$f/ImgH$	2.48
f/R1	3.58	$Y11/ImgH$	0.57
f/R14	0.71	$BL/ImgH$	0.26
f6/f7	1.02		

Furthermore, in the 11th embodiment, among the first lens element **1110**, the second lens element **1120**, the third lens element **1130**, the fourth lens element **1140**, the fifth lens element **1150**, the sixth lens element **1160** and the seventh lens element **1170**, at least one lens element with positive refractive power has an Abbe number which is smaller than 25. That is, the Abbe number of the fifth lens element **1150** is smaller than 25.

12th Embodiment

FIG. 26 is a schematic view of an electronic device **10** according to the 12th embodiment of the present disclosure. The electronic device **10** of the 12th embodiment is a smartphone, wherein the electronic device **10** includes an image capturing device **11**. The image capturing device **11** includes a photographing optical lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the photographing optical lens assembly.

13th Embodiment

FIG. 27 is a schematic view of an electronic device **20** according to the 13th embodiment of the present disclosure. The electronic device **20** of the 13th embodiment is a tablet personal computer, wherein the electronic device **20** includes an image capturing device **21**. The image capturing device **21** includes a photographing optical lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the photographing optical lens assembly.

14th Embodiment

FIG. 28 is a schematic view of an electronic device **30** according to the 14th embodiment of the present disclosure. The electronic device **30** of the 14th embodiment is a wearable device, wherein the electronic device **30** includes an image capturing device **31**. The image capturing device **31** includes a photographing optical lens assembly (its reference numeral is omitted) according to the present

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disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the photographing optical lens assembly.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that Tables 1-22 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A photographing optical lens assembly comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element having negative refractive power;

a third lens element;

a fourth lens element;

a fifth lens element;

a sixth lens element having at least one of an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element comprises at least one inflection point; and

a seventh lens element having an object-side surface and an image-side surface being both aspheric;

wherein the photographing optical lens assembly has a total of seven lens elements, a focal length of the photographing optical lens assembly is f , a curvature radius of the object-side surface of the first lens element is $R1$, a focal length of the sixth lens element is $f6$, a focal length of the seventh lens element is $f7$, a maximal image height of the photographing optical lens assembly is $ImgH$, a maximal optical effective radius of the object-side surface of the first lens element is $Y11$, a curvature radius in a center of the image-side surface of the seventh lens element is $R14$, and the following conditions are satisfied:

$$2.85 < f/R1;$$

$$-2.0 < f6/f7 < 1.5;$$

$$2.20 < f/ImgH < 5.50;$$

$$0.45 < Y11/ImgH < 1.0; \text{ and}$$

$$f/R14 \leq 0.71.$$

2. The photographing optical lens assembly of claim 1, wherein the sixth lens element with negative refractive power has the image-side surface being concave in a paraxial region thereof.

3. The photographing optical lens assembly of claim 1, wherein the second lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, and the fourth lens element has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof.

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4. The photographing optical lens assembly of claim 1, wherein the focal length of the sixth lens element is f_6 , the focal length of the seventh lens element is f_7 , and the following condition is satisfied:

$$-0.60 < f_6/f_7 < 0.60.$$

5. The photographing optical lens assembly of claim 4, wherein the focal length of the sixth lens element is f_6 , the focal length of the seventh lens element is f_7 , and the following condition is satisfied:

$$-0.40 < f_6/f_7 < 0.40.$$

6. The photographing optical lens assembly of claim 1, wherein a refractive power of the first lens element is P_1 , a refractive power of the second lens element is P_2 , a refractive power of the third lens element is P_3 , a refractive power of the fourth lens element is P_4 , a refractive power of the fifth lens element is P_5 , a refractive power of the sixth lens element is P_6 , a refractive power of the seventh lens element is P_7 , and the following condition is satisfied:

$$(|P_3| + |P_4| + |P_5| + |P_7|) / (|P_1| + |P_2| + |P_6|) < 0.50.$$

7. The photographing optical lens assembly of claim 1, wherein an Abbe number of the seventh lens element is V_7 , and the following condition is satisfied:

$$V_7 < 30.$$

8. The photographing optical lens assembly of claim 1, wherein the focal length of the photographing optical lens assembly is f , the curvature radius of the object-side surface of the first lens element is R_1 , and the following condition is satisfied:

$$3.10 < f/R_1 < 7.50.$$

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9. The photographing optical lens assembly of claim 1, further comprising:

an aperture stop disposed between an imaged object and the first lens element, wherein an axial distance between the object-side surface of the first lens element and the aperture stop is $Dr1s$, a central thickness of the second lens element is $CT2$, and the following condition is satisfied:

$$2.0 < |Dr1s|/CT2 < 5.0.$$

10. The photographing optical lens assembly of claim 1, wherein a vertical distance between a non-axial critical point on the image-side surface of the sixth lens element and an optical axis is $Yc62$, a central thickness of the sixth lens element is $CT6$, and the following condition is satisfied:

$$0.5 < Yc62/CT6 < 7.5.$$

11. The photographing optical lens assembly of claim 1, wherein an axial distance between the image-side surface of the seventh lens element and an image surface is BL , the maximal image height of the photographing optical lens assembly is $ImgH$, and the following condition is satisfied:

$$0.10 < BL/ImgH < 0.40.$$

12. An image capturing device, comprising: the photographing optical lens assembly of claim 1; and an image sensor, wherein the image sensor is disposed on an image surface of the photographing optical lens assembly.

13. An electronic device, comprising: the image capturing device of claim 12.

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